

KCSE MOCKS

ALL SUBJECTS

SET 2

Dear Candidates, Attempt these KCSE Mocks!
For Marking Schemes Call 0705525657

443/1

AGRICULTURE

PAPER 1

TIME: 2 HOURS

SECTION A (30 MARKS)

Answer ALL Questions in the Spaces Provided

1. State **two** factors that determine the colour of soil during soil formation. (1mk)

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2. State **four** harmful effects of crop pests in maize production. (2mks)

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3. Give **two** properties of phosphatic fertilizers. (1mk)

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4. List **three** physical agents of chemical weathering. (1½mk)

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5. Mention **four** factors that influence the choice of implement to use in primary cultivation. (2mks)
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6. State **four** reasons of cutting back in pyrethrum. (2mks)
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7. State **two** methods of table formation in tea production. (1mk)
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8. Name the chemical used to achieve the following during water treatment.
- (a) Coagulation of solid particles. (1mk)
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- (b) Softening of water. (1mk)
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- (c) Killing pathogens. (1mk)
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9. Mention **three** methods of land reforms practiced in Kenya. (1½mks)

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10. Distinguish between contact and systemic herbicides in weed control. (1mk)

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11. List **four** biotic factors influencing Agriculture. (2mks)

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12. State **two** methods of farming. (1mk)

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13. Give **four** advantages of title deed to a farmer. (2mks)

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14. State **one** way in which inorganic mulch helps to conserve soil water. (1mk)

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15. Give the meaning of the following terms as used in crop production.

(a) Closed season (½mk)

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(b) Earthing up (½mk)

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16. State **four** benefits of soil organisms in crop growth. (2mks)

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17. State **four** ways in which seeds can be treated before planting. (2mks)

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18. Give **two** ways of forage utilization. (1mk)

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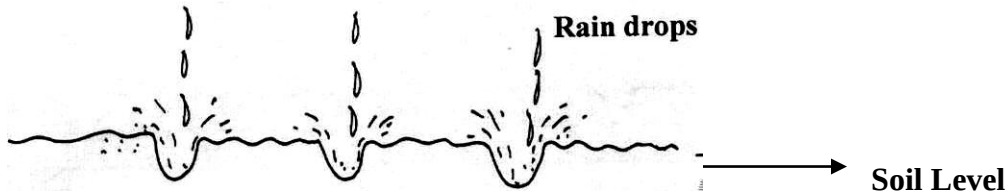
19. Distinguish between scarcity and opportunity cost as used in Agriculture economics (2mks)

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SECTION B (20 MARKS)

Answer all questions in the spaces provided in this section

20. The diagram below illustrates a type of soil erosion. Study it carefully and answer the questions that follow.



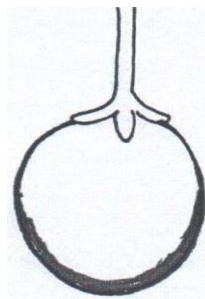
- (a) Identify the type of erosion illustrated above. (1mk)

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- (b) State **two** measures that could be used to control the type of soil erosion illustrated above. (2mks)

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21. The diagram below illustrates a crop disease. Study it carefully and answer the questions that follow.



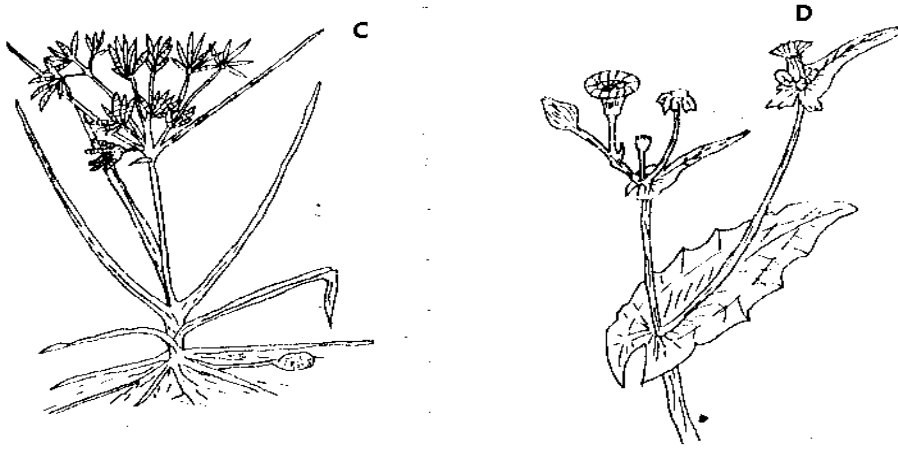
- (a) Identify the disease illustrated above. (1mk)

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- (b) State **three** control measures for the disease illustrated above. (3mks)

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22. Below are crop weeds



a) Identify weed ;

C-

D-

(2mks)

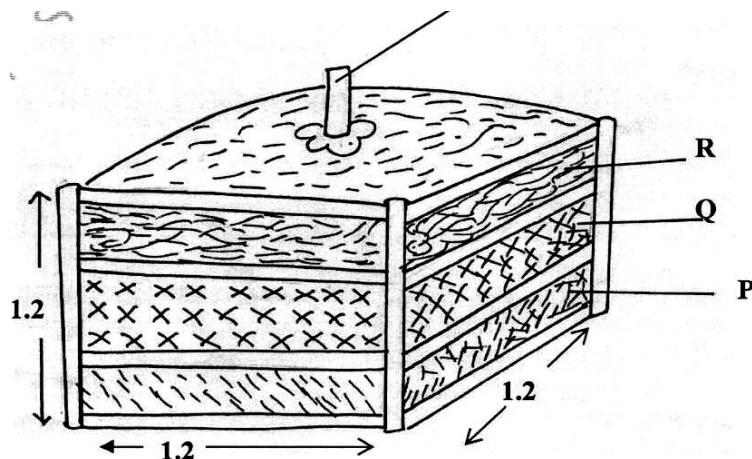
b) Why is it difficult to control weed C

(1mk)

c) State two economic importance of weed D

(2mks)

23. The diagram below illustrates the preparation of compost manure. Study it carefully and answer the questions that follow. T



- (a) Identify the layers labeled P, Q and R above. (3mks)

P:.....

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Q:.....

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R:.....

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- (b) State **one** use of each of the layers labeled P and T. (2mks)

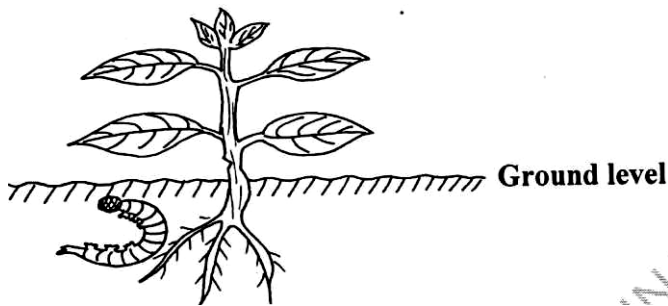
P:.....

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T:.....

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24. Below is an illustration showing a cabbage seedling attacked by a pest. Study it carefully and answer the questions that follow.



- (a) Identify the pest illustrated above. (1mk)

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- (b) State how the pest causes damage to the crop. (1mk)

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- (c) Give **one** method of controlling the pest illustrated above. (1mk)

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SECTION C

Answer any two questions in the spaces provided

25. (a) Explain **five** physical methods that can be used to control pests in the farm(10mks)
(b) Describe five importance of drainage as a method of land reclamation. (10mks)
26. (a) Describe the preparation of silage in a farm. (10mks)
(b) Explain the methods of establishing pastures. (3mks)
(c) Describe the management of pastures. (3mks)
(d) Explain methods of draining water in water logged / marshy areas. (4mks)
27. (a) Explain the uses of water on a farm. (8mks)
(b) Explain reasons for pruning in crops. (7mks)
(c) Describe the procedure of sowing vegetable seeds in a nursery. (5mks)

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KCSE MOCKS SET 2

443/2

AGRICULTURE

PAPER 2

TIME: 2 HRS.

SECTION A (30 MARKS)

Answer ALL Questions in the Spaces Provided

1. Differentiate between heterosis and epistasis (1mk)

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2. Outline **four** possible behaviours of chicks in a brooder when the temperatures are high. (2mks)

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3. A school farm has a deep litter poultry house measuring 9m x 3.5m. If the space allocated per bird is 0.27m^2 , how many birds can the school keep? (1mk)

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4. Name **two** tools that are use for checking the straightness of a wall when laying bricks during construction. (1mk)

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5. State **four** advantages of bucket-feeding a calf. (2mks)

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6. Outline **four** factors that should be considered when selecting dairy goats for breeding. (2mks)

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7. Outline **four** causes of infertility in dairy cattle. (2mks)

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8. Name **four** breeds of rabbits. (2mks)

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9. Give a set of tools that are used for bloodless castration of male calves. (2mks)

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10. State **four** advantages of a spray race over a plunge dip. (2mks)

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11. Name **two** methods that are used in selection of cattle for breeding.

(1mk)

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12. Give **four** symptoms of calf scours.

(2mks)

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13. Give two reasons for treating timber (1mk)

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14. Name the part of the digestive system of poultry which acts as the true stomach.

(1mk)

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15. Outline **four** factors that determine the nutritional requirements of an animal.

(2mks)

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16. Give the uses of the named tools .

(2mks)

- a) Wire strainer-
- b) Trocar and canula-
- c) Soil auger-
- d) Scalpel-

17. List **four** reasons why livestock should be kept healthy.

(2mks)

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18. State **four** harmful effects of ticks on livestock.

(2mks)

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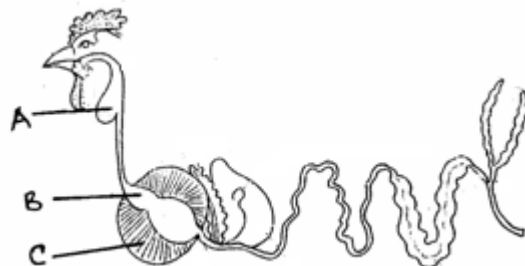
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SECTION B :(20MARKS)

Answer all questions from this section.

19. The following is a diagram showing part of the poultry's digestive system. Use it to answer the question that follow.



- (i) Name the parts labelled **A**, **B**, **C**. (3mks)

A-

B-

C-

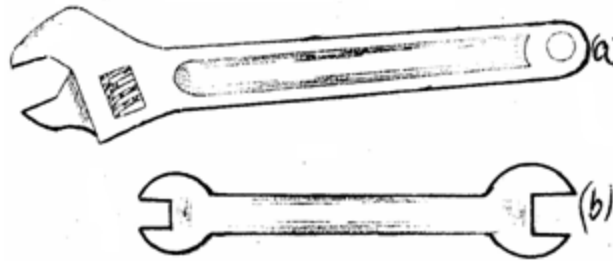
- (ii) Give **two** structural adaptations of **C** to its functions. (2mks)

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20. The diagram below shows examples of workshop tools and equipment.



- (i) Identify tool **(a)**.....(1mk)
- (ii) Identify tool **(b)**.....(1mk)
- (iii)
- (b) State **one** advantage of tool **a** over tool **b**. (1mk)
- (c) State the maintenance practices of the above tools. (2mks)

21.The following illustration represents a certain practice carried out in sheep management. Study the illustration and answer the questions that follow.



(i) Identify the practice.

(1mk)

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(ii) Name the tool being used in carrying out the practice.

(1mk)

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.....

(iii) State **two** precautions that should be taken when carrying out this practice.

(2mks)

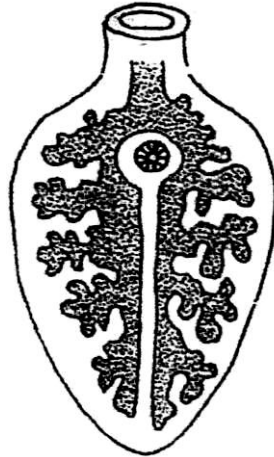
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(iv) How often should the practice be carried out.

(1mk)

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22.Below is an illustration of an internal parasite in livestock.



(a) Identify the parasite.

(1mk)

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(b) Name **one** final host for the parasite.

(1mk)

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(c) Name the intermediate host for the parasite.

(1mk)

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(d) State **two** measures used to control the parasite.

(2mks)

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SECTION C (40MKS)

Answer any two questions in this section

23. Describe the management practices carried out on ewes from preparation for mating to weaning of lambs. (20mks)

24.(a) Describe the life cycle of a two-host tick. (8mks)

(b) Discuss Newcastle disease under the following sub-headings:-

i)Mode of transmission. (2mks)

ii)Symptoms of attack. (5mks)

iii)Control measures. (5mks)

25.(a) What are the characteristics of dairy cattle? (4mks)

(b) Describe the procedure followed when constructing a barbed wire fence.
(8mks)

(c) Describe the digestion of grass in the rumen of a ruminant.
(8mks)

KCSE MOCKS SET 2

BIOLOGY 232/1

Answer all questions

1. Name the structures used for locomotion in each of the following organism

(a) Euglena (1 mark)

(b) Paramecium (1 mark)

2. (a) What is sex - linkage? (2 marks)

(b) Name one sex- linked trait in human beings (1 mark)

3. Blackjack (bidens pilosa) belongs to the family compositae. What is the plants

(a) Genus (1 mark)

(b) Species (1 mark)

4. Name two metabolic waste products in

(a) Birds (2 mark)

(b) Plants (2 marks)

5. State the adaptations of seed to dispersal by wind (3 marks)

6. State the importance of the growth of pollen tubes in flowering plants (1 mark)

7. State three structural differences between DNA and RNA in living cells (3 marks)

8. (a) State two differences between meiosis and mitosis (2 marks)

(b) State two processes that takes place during interphase (2 marks)

9. Name two parts in the human body with cilia (2 marks)

10. The diagram below represents a closed stoma



(a) Identify the cells labeled A and B (2 marks)

(b) Name the excretory product in plants which is excreted through the stomata

(1 mark)

(c) State one adaptation of the guard cell to its function (1 mark)

11. Name two organisms that form the biological environment of a malaria parasite

(2 marks)

12. Name the organs of the mammalian body that are responsible for production of gametes

(2 marks)

13. List three adaptations of fruits that are dispersed by animals (3 marks)

14. The equation below show what happens in cellular respiration



(a) Name the type of respiration shown and where it occurs in a cell

(2 marks)

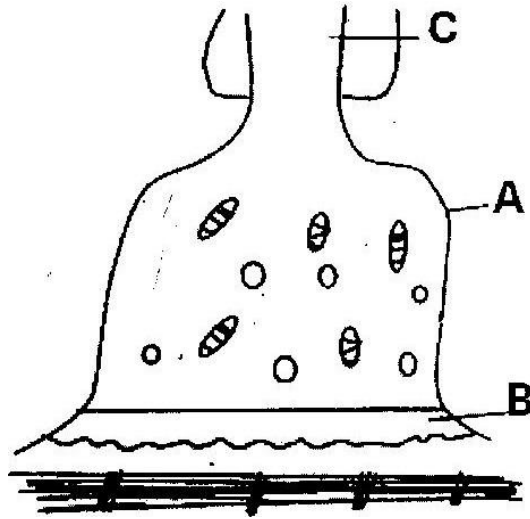
(b) Determine the respiratory quotient of the process (2 marks)

(c) What food substrate is broken in the respiration? (1 mark)

15. List two features of the small intestine that increase its surface area

(2 marks)

16.



The diagram above shows synapse at a neuromuscular junction

- (a) Name the parts labeled A and B
 - (b) State the function of the part labeled C
17. Explain why food is stored in an insoluble form in the cells of living things
- (2 marks)
18. State the observation made in germinating seeds when the
- (a) Hypocotyle elongates
 - (b) Epicotyle elongates

19. State the differences between assimilation and absorption of food nutrients

(2 marks)

20. State three homeostatic function of the liver

21. Differentiate between lactic fermentation and alcoholic fermentation

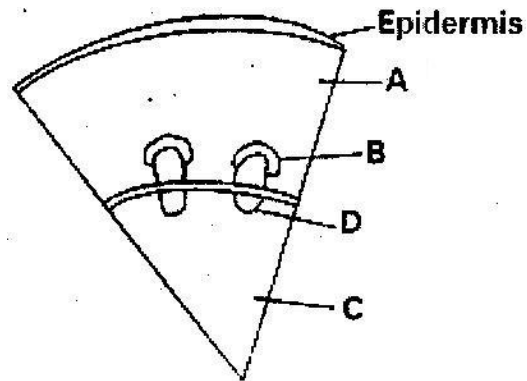
(4 marks)

22. Water logging can cause the death of some plants. Explain (2 marks)

23. Distinguish between plasmolysis and haemolysis (2 marks)

24. John and grace who are siblings are both normal and so are their parents but they have a haemophilic brother. Give the genotypes of their parents. (2 marks)

25. The diagram below shows a section of a dicotyledonous stem.



Name the tissues labeled A and D and state the function of each.

26. Name the organism that causes each of the following diseases

(a) AIDS

(b) Bilhazia

(c) Cholera

(3 marks)

27. List three examples of gaseous exchanges surfaces in animals (3 marks)

28. state the significance of photosynthesis (3 marks)

29. Explain the meaning of each of the following

(a) Continental drift

(b) Fossils (2 marks)

30. Green plants grow towards a source of light

Name this type of response (1 mark)

KCSE MOCKS SET 2

BIOLOGY 231/2

TIME 2 HOURS

This paper consists A and B. Answer ALL the question in section A in the spaces provided. In Section B answer question 6 (compulsory) and either question 7 or 8 in the spaces provided.

SECTION A (40 MARKS)

Answer all the questions in this section

1. (a) What is gene linkage (1 mark)

(b) Haemophilia is sex linked trait

(i) If a normal woman but a carrier for haemophilia marries a normal man work

Out the phenotype of the offspring using a genetic cross. (5 marks)

(ii) Why is haemophilia, more common defect in males than in females?

(1 mark)

(ii) Other than haemophilia state any other sex linked defect in man (1 mark)

2. The table below shows the percentage composition by volume of inhaled and exhaled air

Gas	Inhales air %	Exhales air %
Oxygen	21	16
Carbon (IV) Oxide	0.04	4.0
Nitrogen	79	79

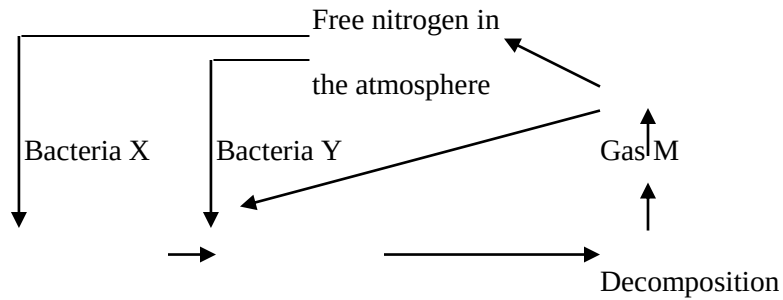
- (a) (i) By what percentage is the carbon (IV) Oxide concentration in exhaled air higher than inhaled air? (3 marks)

- (ii) Explain the differences in the composition of the gases between inhaled and exhaled air. (3 marks)

- (b) State two ways in which leaves of plant are adapted for gaseous exchange?

(2 marks)

3. The chart below represents the flow of nitrogen in the ecosystem



Nitrates in the soil **A** Plant protein **B** Animal protein

(a) Name the bacteria labeled X and Y (2 marks)

(b) Name the gas M

(c) Name processes A, B and C

(c) State the bacteria involved in the process named C

4. An experiment was set up to investigate the effect of unilateral light in growth of maize seedlings.

The diagram in the table below represents experimental set up at the begin

Experimental set up	Beginning of experiment	Expected results
C		

D		
B		

(a) Using diagrams complete the table to show the expected results in experimental set up.

(3 marks)

(b) Account for your results in experimental set up (3 marks)

(c) Explain the purpose of experimental set up B and C (2 marks)

5. (a) What is internal fertilization? (1 mark)

(b) Suggest two disadvantages of internal fertilization in most mammals
(2 marks)

(c) State two roles of placenta in mammals (2 marks)

(d) Mention one role played by each of the following hormones in human

Menstrual cycle

(a) Follicle stimulating hormones (FSH)

(ii) Oestrogen

(iii) Luteinizing Hormone (LH)

SECTION B (40 MARKS)

Answer question (compulsory) in the spaces provided either question 7 & 8 in the spaces provided after question 8.

6. The following data are results of making daily growth measurement ion an organism

Over a period of 24 days during its development.

Day	Width of head (mm)	Length of hind femur (mm)
1.	3.0	7.0
2.		
3.	3.5	7.5
4.	4.0	8.0
5.		
6.	4.0	8.0
7.	4.0	8.0
8.		
9.	4.0	9.2
10.	4.0	10.5
11.		
12.	4.4	12.0
13.		
14.	4.7	12.0
15.	5.0	12.0
16.		
17.	5.0	12.0
18.	5.0	12.0
19.		
20.	5.0	12.0
21.	5.0	12.0
22.		
23.	5.0	13.3
24.	5.0	14.8
	5.7	16.4
	6.4	18.0
	7.0	18.0
	7.6	18.0
	7.6	18.0
	7.6	18.0
	7.6	18.0
	7.6	18.0

- (a) Using a suitable scale draw graphs of width of head and length of femur against time.

Draw the graphs on the same axis. (8 marks)

- (b) (i) Name the growth pattern represented by the graph (1 mark)

- (ii) With reference to your graph identify the phylum to which the organisms belong.

Give a reason for your answer (2 marks)

- (c) Account for the length of hind femur between

- (i) Day 3 and day 7 (3 marks)

- (ii) Day 7 and day 10 (2 marks)

- (d) State two hormones involved in the growth pattern represented by the graphs

(2 marks)

- (e) State two advantages of metamorphosis in organisms (2 marks)

7. Describe how water and mineral salts move from soil until they reach the leaves in a tall plant. (20 marks)
8. (a) Describe the following terms:
- (i) Secretion
 - (ii) Excretion
 - (iii) Egestion (3 marks)
- (d) Explain how the mammalian kidney is adapted to its functions. (17 marks)

KCSE MOCKS SET 2

Business Studies

Paper 1

Time: 2Hours

QUESTIONS

1. (a) Outline **four** benefits that a business gain from its environment.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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2. Highlight **four** limitations that small scale retailer operators face.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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3. Give **four** factors that influence the location of a public warehouse.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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4. Outline **four** ways in which inflation can be controlled by the government.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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5. State **four** circumstances that would make the volume of stock change in a business.
(4mrks)

(i)

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(ii)

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(iii)

.....

(iv)

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6. Give **four** circumstances when a country may be forced to control her imports.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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7. Highlight **four** uses of National Income statistics.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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8. State **four** challenges that a business with no plan will encounter in its operations.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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9. Highlight **four** merits of indirect tax in the Kenyan economy.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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10. State **four** services that the central bank of Kenya offers to commercial banks.
(4mrks)

(i)

.....

(ii)

(iii)

(iv)

11. Outline **four** strategies that can be used by small scale firms in order to expand.
(4mrks)

(i)

(ii)

(iii)

(iv)

12. Highlight any **four** contemporary trends of communication in Kenya today.
(4mrks)

(i)

(ii)

(iii)

(iv)

13. Highlight **four** reasons why the owner of a small business enterprise would prefer to advertise his goods through internet.
(4mrks)

(i)

(ii)

(iii)

(iv)

14. Outline **four** roles played by Licensing as a government regulatory tool.
(4mrks)

(i)

(ii)

(iii)

(iv)

15. State **four** reasons why land is considered a prime factor of production.
(4mrks)

(i)

(ii)

(iii)

(iv)

16. State **four** benefits of replacing manual system with automatic one in carrying out office activities.
(4mrks)

(i)

(ii)

(iii)

(iv)

17. Outline **four** features of a general or property Insurance policy.
(4mrks)

(i)

(ii)

(iii)

(iv)

18. Outline **four** differences between the liners and tramps in sea transport.

LINERS	TRAMPS
(i)	
(ii)	
(iii)	
(iv)	

19. Outline **four** measures being used by the government to control high rate of unemployment in Kenya. (4mrks)

(i)

(ii)

(iii)

(iii)

(iv)

20. For each of the following transactions indicates in the space provided the type of journal in which the record would be made.

(4mrks)

Transaction	Journal
(i) Motor vehicle bought on credit	
(ii) Bought goods on credit	
(iii) Returned goods to credit suppliers	
(iv) Sold goods on credit	

21. The following balance were extracted from the books of Kiboko Traders on 31st January 2014.

SHS.

Carriage outwards	1,800
Capital	10,000
Commission allowed	2,500
Drawing	1,500
Discount received	1,750
Furniture	8,200
Commission received	1,200
Creditors	1,050

Required: Prepare Trial balance as at 31 January 2014.

22. Outline **four** indicators of economic development.
(4mrks)

(i)

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(ii)

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(iii)

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(iv)

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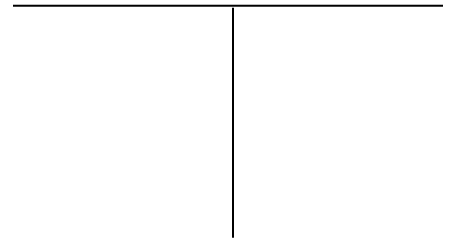
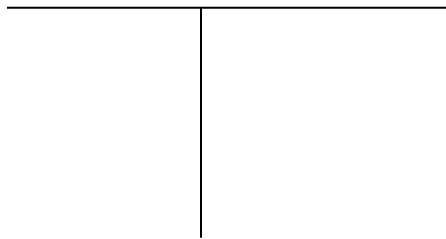
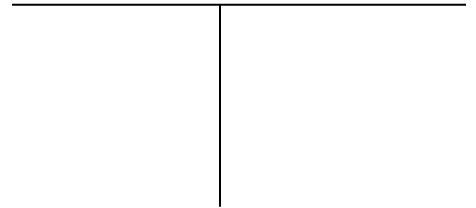
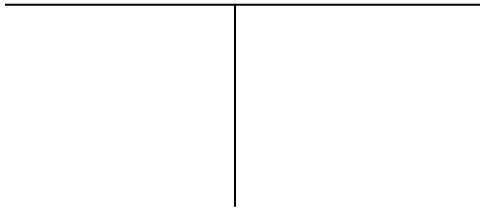
23. The following information were extracted from books of Jumbo Traders on 31st December 2013.

SHS.

General expenses
Discount allowed
Discount received
Gross profit
Salaries
Rent payment
Electricity

- Required: Repair a profit and Loss account for Jumbo Traders for the period ended 31st December
2013

24. The following transaction relate to the Masumbuko traders on 1st January 2014.
- (i) Started business with shs. 250,000 cash in Bank.
 - (ii) Bought goods for shs. 200,000 from Otieno on credit.
- (4mrk)



25. Nyayo Traders had the following during the year 2013.

	SHS.
Capita 1- 1- 2013	420,000
Net profit	70,000
Drawing	130,000
Capital 31-12-2013	500,000

Required: Find the value of investment?

KCSE MOCKS SET 2

565/2

BUSINESS STUDIES

Paper 2

Time: 2 ½ Hours

- 1) a) Bidii Nyakio is a large-scale retailer shops in Nairobi, Machakos, Nyeri and Nakuru. Explain
Five benefits of operating such shops. (10Marks)
- b) Outline Five principles of taxation. (10 Marks)
- 2) a) Buildings in business centers are constructed in a way to display the goods on sale. Explain
Five disadvantages of using that technique of sales-promotion. (10 Marks)
- b) Explain Five factors that account for the existence of small firms. (10 Marks)
- 3a) Explain Five factors that retard development in most developing nations? (10 Marks)
- b) Explain Five types of unemployment in developing countries. (10 Marks)
4. (a) Record the following transactions in Muema's three column cashbook and balance it off as at 30th June 2012. (12 marks)
- 2012**
- June 2. KCB loan received by cheque Sh. 250,000
- June 4. Cash sales amounting to Sh. 30,000.
- June 6. Bought stock worth Sh.70,000 and paid by cheque less 8% discount.
- June 9. Received sh. 25,000 from Mr. Lonyangapuo a debtor by cheque after allowing 2% cash discount.
- June 10. The proprietor took Sh.10,000 cash for his son's birthday party.
- June 12. Withdrew Sh.48,000 from the bank for office use.
- June 15. Murgor settled his account of shs. 45,000 by cheque.
- June 18. Paid wages Sh.10,000 Cash.
- June 20. Murgor's cheque was dishonoured.
- June 27. Sold goods worth Sh. 68,000 receiving Sh.28,000 cash and the rest by cheque.
- June 30. Transferred all cash to the bank Except Sh.8,000.

(b) Explain **Four** ways through which the central bank control inflation. (8 marks)

5(a) Explain **Five** Measures to safeguard the property of an organization (10 marks)

(b) Explain **Five** Reasons that account for popularity of SACCOs in Kenya (10 marks)

6(a) The following Trial Balance was extracted from the books of Umoja General Merchants on 31 December 2013. (10 Marks)

Umoja General Merchants
Trial Balance as at 31 December 2013

	Dr.	Cr.
Motor vehicles	322,500	
Purchases	157,500	
Sales		337,500
Premises	412,500	
Drawings	7,500	
Capital		741,750
Bank		9,750
Cash	48,000	
Debtors	52,500	
Creditors		22,500
Stock	30,000	
Discounts	7,500	3,000
Returns	18,750	23,250
Salaries	75,000	
Power and Lighting	6,000	
	1,137,750	1,137,750

Stock as at 31 December 2013 was valued at Sh.16,500.

Required:

- i) Prepare the Trading and Profit and Loss Account for the year ending 31 December 2013. (7 Marks)
- ii) Balance Sheet as at 31 December 2013. (5 Marks)
- b) Explain **Four** reasons why entrepreneurship is important to an economy. (8 Marks)

KCSE MOCKS SET 2

C.R.E. PAPER 1 (313/1)

TIME: 2¹/₂ HOURS

INSTRUCTIONS:

Answer any Five Questions:

1.

- (a) Outline the difference in the two accounts of creation in Genesis 1 and 2 (10 mks)
- (b) From the story of the fall of human beings in Genesis Chapter 3, state the effects of sin (6 mks)
- (c) Identify any four cases of evil in Kenya today (4 mks)

2.

- (a) Describe how God prepared the Israelites for the making of the covenant at Mt. Sinai (Exodus 19). (8 marks)
- (b) Give four reasons that made the Israelites to break the covenant while at Mt. Sinai (Exodus 32: 1-35). (4 marks)
- (c) What do Christians learn about the nature of God from the Exodus? (8 marks)

3.

- (a) State seven functions of the temple in The Jewish community (7 mks)
- (b) Identify six ways which show that king Solomon turned away from the covenant way of life (6 mks)
- (c) Give seven factors that have led to the increase of Christian denominations in Kenya today (7 mks)

4.

- (a) Identify six characteristics of false prophets in the Old Testament. (6 marks)
- (b) Describe the teaching of prophet Amos on Israel's election. (6 marks)
- (c) State eight factors that hinder Christians from practicing their faith in God. (8 marks)

5.

- (a) Describe the political background to Nehemiah. (5 marks)
- (b) Identify seven occasions when Nehemiah prayed. (7 marks)
- (c) State the importance of prayer in the life of a Christian. (8 marks)

6.

- (a) Describe five ways in which people in traditional African communities communicate with God. (5 marks)
- (b) Give eight reasons for singing and dancing during initiation ceremonies in traditional African communities. (8 marks)
- (c) Give seven reasons why witchcraft is feared in traditional African communities. (7 marks)

KCSE MOCKS SET 2

C.R.E. PAPER 2 (313/2)

TIME: 2¹/₂ HOURS

INSTRUCTIONS:

Answer any Five Questions

(a) Basing your answer on the infancy narratives in Luke 1:5-56, describe what took place when Mary visited Elizabeth. (6 marks)

(b) Identify six lessons that Christians learn from the lives of Zechariah and Elizabeth. (6 marks)

(c) State eight ways through which Christians in Kenya express their joy for the birth of Jesus. (8 marks)

2.

(a) Describe the call of the first disciples of Jesus. (Luke 5:1-11). (8 marks)

(b) Give six reasons why Jesus chose the twelve disciples. (6 marks)

(c) Identify six lessons that Christians learn from the call of the first disciples. (6 marks)

3 (a) Describe the triumphant entry of Jesus into Jerusalem. (Luke 19:28-40) (6 marks)

(b) Outline the events that took place from the time Jesus was arrested up to the time he was sentenced to die. (7 marks)

(c) State seven lessons Christians learn from the suffering and death of Jesus. (7 marks)

4. (a) Explain the teaching of Peter concerning the people of God (1st Peter 2: 9 - 10). (10 marks)

(b) Give six ways through which Christians can promote unity among themselves in Kenya today.

(6 marks)

(c) State how kindness as a fruit of the Holy Spirit is abused in the Church in Kenya today. (4 marks)

5. (a) Outline the Christian teachings on marriage. (6 marks)

(b) Give six reasons why some young people are choosing to remain unmarried in Kenya today.

(8 marks)

(c) Identify six ways in which the Church is helping to solve the problem of domestic violence in Kenya today. (6 marks)

6.(a) Explain how science and technology has improved human life. (8 marks)

(b) Give six reasons why Christians are opposed to euthanasia. (6 marks)

(c) Identify ways through which Christians can help to control desertification. (6 marks)

KCSE MOCKS SET 2

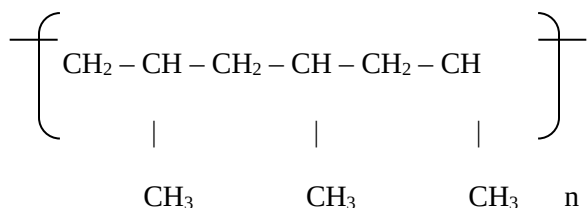
KCSE

CHEMISTRY THEORY

Paper 1

Time: 2 Hours

1. A polymer has the following structure.



- (a) Draw the structure of the monomer. (1mk)
- (b) A sample of the polymer is found to have molecular mass of 6426. Determine the number of monomers present. (C = 12, H = 1). (2mks)
2. Name the catalyst used in:
- (a) the decomposition of hydrogen peroxide. (1mk)
- (b) the conversion of vegetable oil into margarine. (1 mk)

3. (a) Using dots (·) and crosses (x), draw the dimer structure of aluminium chloride and name the bonds. (2mks)
- b) State and explain the observation made when sodium carbonate powder is dropped into a solution of aluminium chloride. (1mks)
4. Determine the heat of solution of potassium fluoride given that its lattice energy is +801KJ/mole and the heats of hydration of potassium and fluoride ions are -322 and -506KJ/mole respectively. (3mks)
5. During the extraction of aluminium, a current of 0.2 ampere was passed for one hour through the molten aluminium oxide.
- (a) Write equations for the reaction that took place at the cathode. (A1 = 27, 1F = 96500C). (1mk)

(b) Calculate the mass of aluminium produced. (2mks)

6(a) Define Gay – Lussac’s law. (1 mk)

(b) 30 cm³ of ethane were mixed with 100 cm³ of oxygen and the mixture was sparked to complete reaction all the volumes were measured at a pressure of one atmosphere and a temperature of 25°C, calculate the volume the residual gas under room temperature. (2 mks)

7. The following table shows the pH values of solutions A, B, C and D.

Solution	A	B	C	D
pH	3	7	10	14

(i) Which solution is likely to be that of iron (III) chloride? (1 mk)

(ii) Which solution has the highest concentration of hydroxide? Explain. (2 mks)

8. The table below gives information of four elements by letters A, B, C and D. study it and answer the questions that follow. The letters do not represent the actual symbol of the elements.

Element	Electron arrangement	Atomic radius (nm)	Ionic radius nm
A	2.8.2	0.136	0.065
B	2.8.7	0.097	0.181
C	2.8.8.1	0.203	0.133
D	2.8.1	0.174	0.097

(a) Which two elements have similar properties? Explain. (2 mks)

(b) Why is the ionic radius of element B greater than its atomic radius? (1 mk)

9. Starting with Lead (II) oxide, describe how you can prepare a sample of lead (II) sulphate. 3mks

10. When hydrogen gas is passed over Lead (II) oxide in a combustion tube, Lead (II) oxide is reduced.

(a) Write an equation for the above reaction. (1 mk)

(b) What observations were made in the combustion tube when the reaction was complete? (2 mk)

11. Explain why potassium is kept in paraffin while phosphorous is kept in water (2mks)

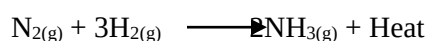
12. Sulphur is soluble in ethanol but not in water while common salt is soluble in water but not in ethanol

a) Explain why sulphur is not soluble in water (1mk)

b) Explain how a pure sample of sodium chloride can be obtained from a mixture of the two

(1mk)

13. Ammonia gas is prepared by Haber's process according to the equation below



Complete the table below by stating the effect of equilibrium when the following conditions are applied. Give explanation in each case

Condition	Effect on equilibrium	Explanation
a) Pressure increased	½ mk	1mk
b) Temperature increased	½ mk	1mk

14. State what would be observed if concentrated sulphuric (VI) acid is added to

a) Sugar crystals.
(1mk)

b) Hydrated copper (II) sulphate solution
(1mk)

c) What type of reaction has taken place above (1mk)

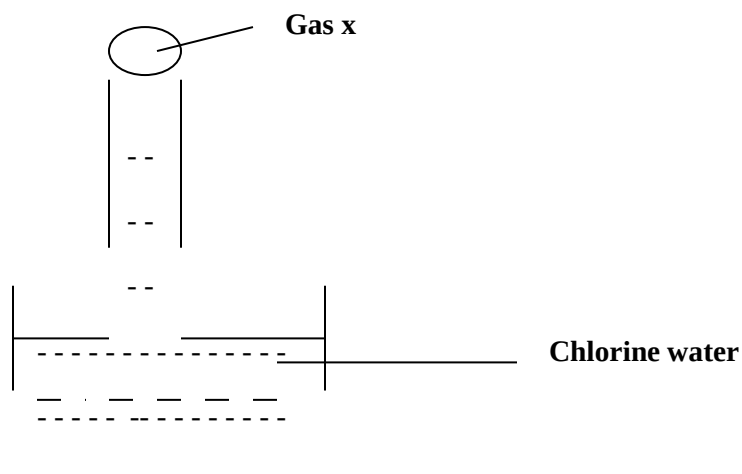
15. Name three sub – atomic particles found in an atom and state where they are found (3mks)

16. The solubility of potassium nitrate is 85g/100g of water at 50°C and 32g/100g of water at 25°C.

a) Define the term solubility. (1mk)

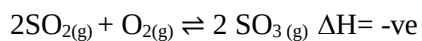
b. Calculate the mass of the crystals formed if a saturated solution of potassium nitrate in 50g of water at 50°C is cooled to 25°C. (2 marks)

17. Study the set-up below and answer the questions that follow.



- a) Name gas **x**..... (1 mark)
- b) State the condition which is not indicated on the diagram for gas **x** to be formed.(1mk)
18. Calculate the mass of sulphur which on complete combustion would yield 7dm^3 of sulphur (iv) oxide measured at 182°C and 722 mm Hg pressure ($O=16$, $S=32$, molar gas volume = 24dm^3 at r.t.p)
(3 marks)
19. a) State and explain the observations made when fluorine gas is bubbled through sodium bromide solution.
(2 marks)
- b) When excess ammonia solution is added to a solution of copper (ii) ions, a deep blue solution forms. Write the formula of the complex ions formed.
(1 mark)

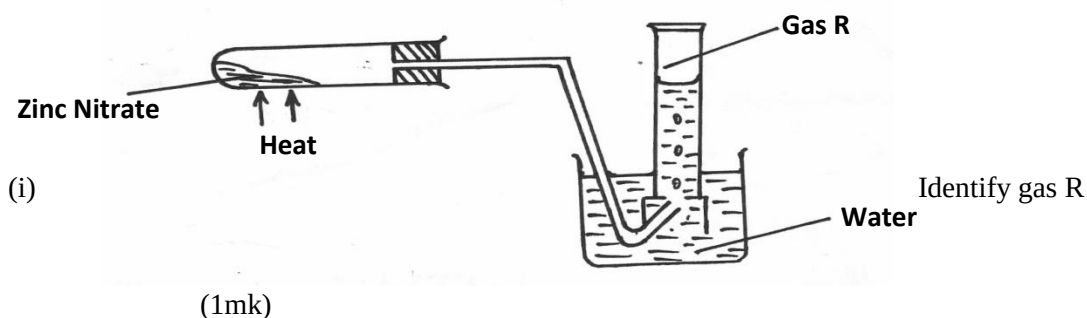
20. The following reaction is in equilibrium in a closed container



State giving reasons how an increase in temperature would affect the amount of sulphur

(VI) oxide gas (2mk)

21. Study the diagram below and use it to answer questions that follows



(i) What changes would be observed in the test-tube (1mk)

(ii) Explain the nature of the solution formed in water after sometimes (2mks)

22. Explain the following observations

(i) electrical conductivity of metals decreases with an increase in temperature (2mks)

(ii) Ionization energy decreases down the group in group I (2mks)

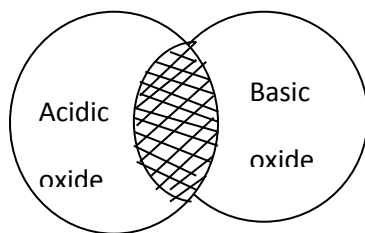
23. M grams of a radioactive isotope decayed to 5 grams in 100 days. The half-life of the isotope is 25 days.

(i) What is meant by half-life (1mk)

(ii) Calculate the initial mass of the radioactive isotope (2mks)

24. Concentrated hydrochloric acid is used for removing oxides from metals (pickling). Explain why concentrated nitric (V) acid cannot be used for the same purpose (2mks)

25. The diagram below shows acidic and basic oxides fit into the general family of oxides



(i) State the name given to the type of oxides that would be placed in the shaded region (1mk)

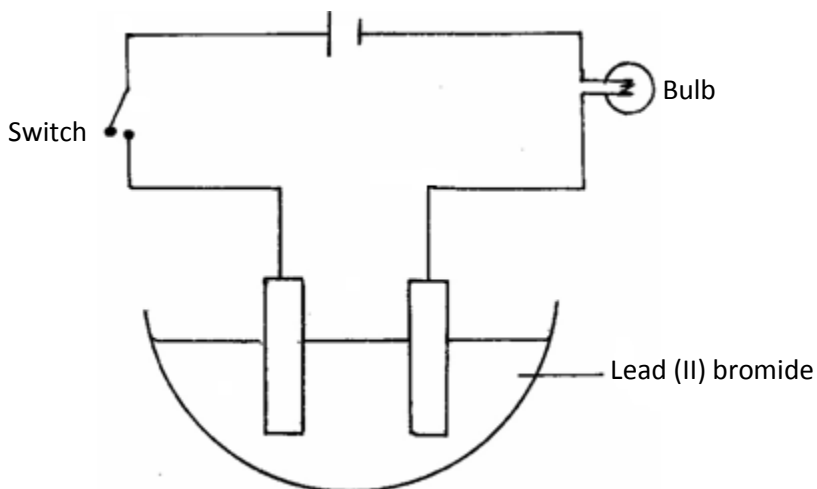
(ii) Give the names of oxides that could be placed in the shaded region (3mk)

26. Explain the following observations.

(a) A beaker with lime water, when left exposed develops a layer of a white solid on the surface. (1 mark)

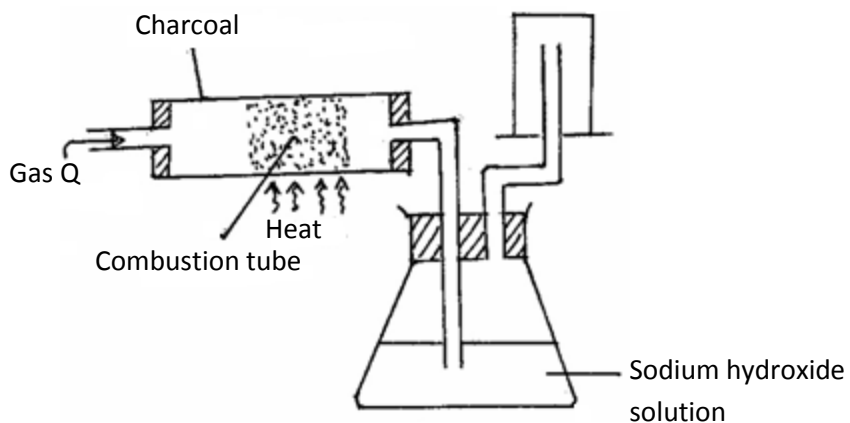
(b) Calcium chloride powder, when left exposed gradually turn into a colourless solution. (1 mark)

27. The diagram **below** shows a set-up intended for the electrolysis of molten lead (II) bromide.



- (a) Label on the diagram: (i) anode (½ mark)
(ii) cathode (½ mark)
- (b) Indicate on the diagram the direction of flow of electrons. (1 mark)
- (c) State the observations made at the
- (i) anode (½ mark)
- (ii) cathode (½ mark)

28. The diagram **below** shows an experimental set-up for preparing carbon (II) oxide. Study it and answer the questions that follow.

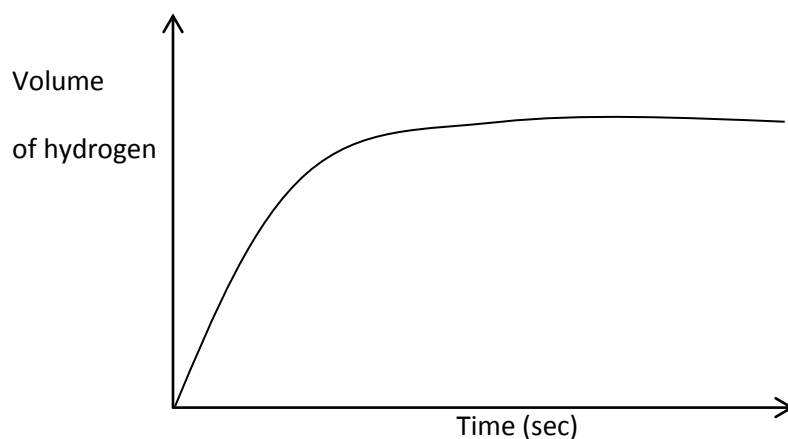


(a) State the role of sodium hydroxide solution in the set-up. (1 mark)

(b) State the reason why carbon (II) oxide is collected in the manner indicated.(1 mark)

(c) Describe a simple test that can be used to distinguish between carbon (II) oxide and carbon (IV) oxide. (1 mark)

29. In an experiment to monitor the rate of reaction of zinc and hydrochloric acid a student recorded the volume of hydrogen produced at regular time intervals and obtained the graph shown **below**.



(a) On the same set of axes sketch the curve expected if the experiment is repeated with a few crystals of copper (II) sulphate added to the reactants. (1 mark)

(b) Explain the shape of your curve. (1 mark)

KCSE MOCKS SET 2

FORM 4

CHEMISTRY THEORY

Paper2

Time: 2 Hours

1. (a) Give the names of the following compounds.

(i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (1 mk)

(ii) $\text{CH}_3\text{CH}_2\text{COOH}$ (1 mk)

- (b) Study the information in the table below and answer the questions that follow.

No. of carbon atoms per molecule	Relative molecular mass of hydrogen
2	28
3	42
4	56

(i) Write the general formula of the hydrocarbons in the table. (1 mk)

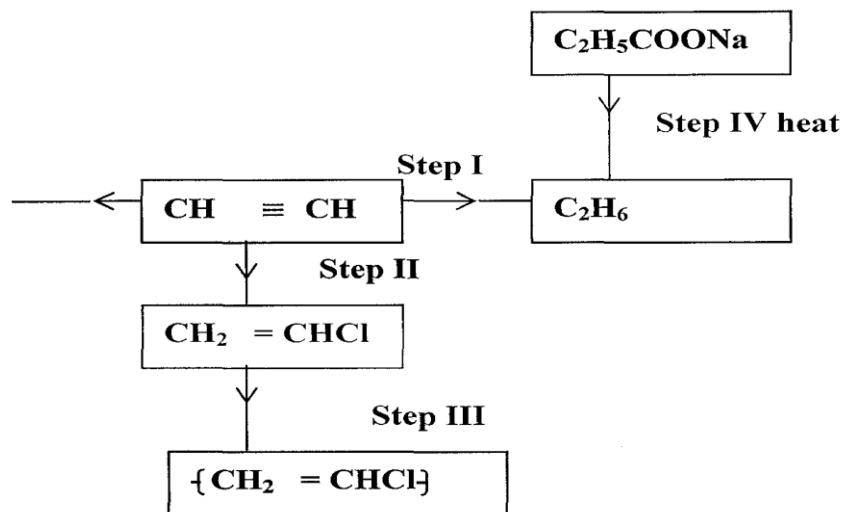
(ii) Predict the relative molecular mass of the hydrocarbon with 5 carbon atoms (1mk)

(iii) Determine the molecular formula of the hydrocarbon in (ii) and draw its structural

formula.

(2 mks)

(c) Study the scheme given below and answer questions that follow.



(i) Name the reagent used in

Step I

(1

mk)

Step II

(1 mk)

Step IV

(1 mk)

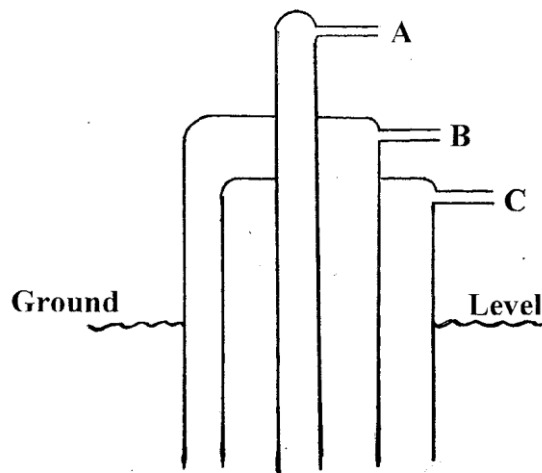
(ii) Write an equation for complete combustion of $\text{CH} \equiv \text{CH}$

(1 mk)

(iii) Explain **one** disadvantage of the continued use of items in step III.

(1 mk)

2. (a) The diagram below represents the extraction of sulphur by the frasch process.



- (i) Identify and state the use of the substances that pass through tubes A and C.

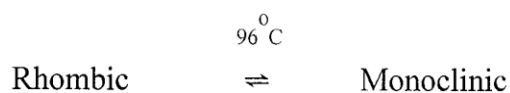
(2

mks)

A.....

C.....

- (ii) Rhombic and monoclinic are Allotropes of sulphur. They are inter convertible as shown below.



- I. What does the temperature 96°C represent.

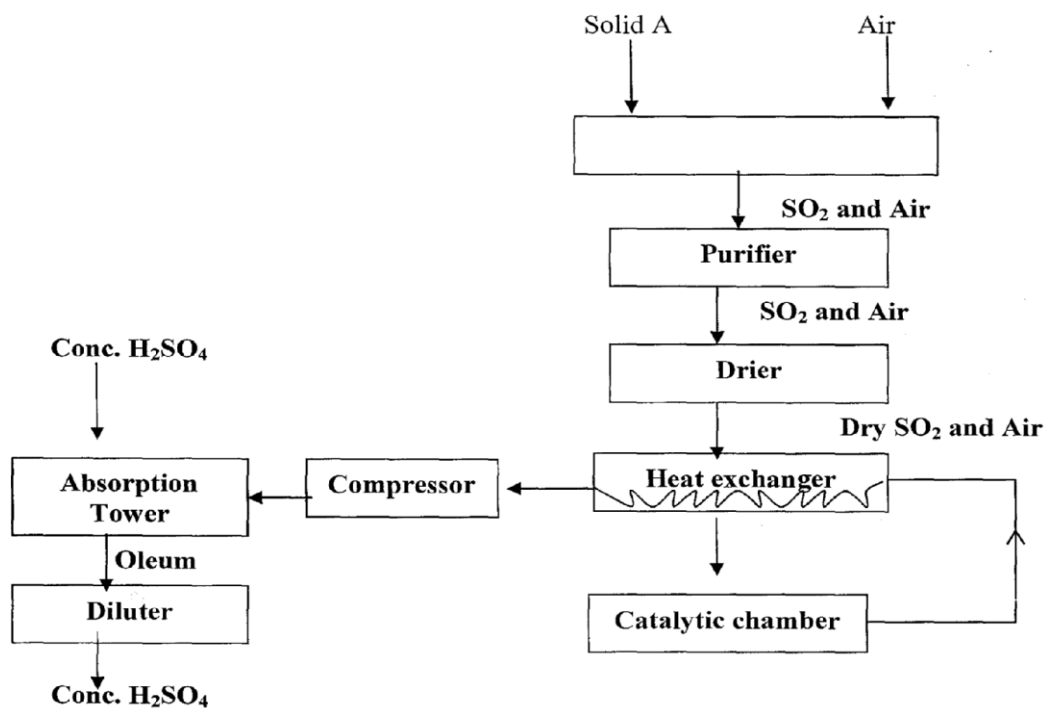
(1

mk)

- II. State the differences in crystalline appearances between rhombic and monoclinic crystals.

(1mk)

- (b) The following scheme represents the steps followed in the contact process, study it and answer the questions which follow.



- (i) Name **three** possible identities of solid A.

(1

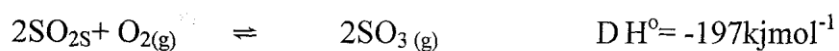
mk)

- (ii) Name **two** impurities removed by the purifier.

(1 mk)

(iii) Why is it necessary to remove impurities? (1 mk)

(c) The following chemical equation shows a reaction taking place in the catalytic chamber/converter.



(i) How would the following factors affect the production of sulphur (IV) oxide. (1 mk)

I Increase in temperature.

(1 mk) II Decrease in pressure

(ii) Name the catalyst which is commonly used in this process and why? (1 mk)

(iii) State and explain one environmental effect of sulphur (IV) oxide in the atmosphere. (2mks)

3. The table below shows some elements of the periodic table and their atomic numbers, atomic masses and melting points. The letters are not the actual symbols of the elements.

Element	B	C	D	E	F	G	H	I	J	K
Atomic No.	7	8	19	15	2	9	6	16	12	11
Atomic mass	14	16	39	31	4	19	12	32	40	23
Melting point $^{\circ}\text{C}$	-209	-218	63.7	44	-272	-223	Vary	113	669	98

(a) Select two elements with oxidation state of -3. (1mk)

(b) Which elements represents the most powerful reducing agent. Explain. (2mks)

(c) How does the atomic radii of D compare with that of K. Explain. (2mks)

(d) How do you compare the electrical conductivity of elements J and K. Give your reason. (1mk)

(e) Select two elements which when reacted with element G form a compound that conducts electricity both in molten and in aqueous state. (2mk)

(f) Explain why the melting point of H has not been given a specific value. (1mk)

(g) In which group and period do D and G belong?

(2mks)

D Group _____

Period _____

G Group _____

Period _____

(h) Explain why the melting point of element K is higher than that of D.

(1mk)

(i) Select one element stored in

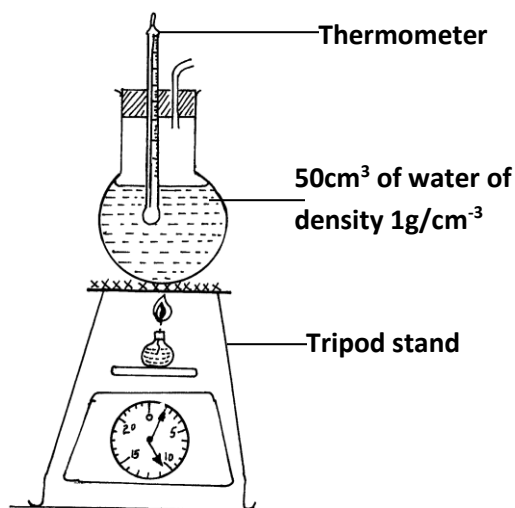
(a) Water _____

(1mk)

(b) Paraffin _____

(1mk)

4. The following set – up was made in an experiment by a group of form four students. The readings of the balance before and after experiment were indicated in the diagram below. Given that the initial temperature of water was 26.7°C respectively. The specific heat capacity of water is $4200\text{Jkg}^{-1}\text{k}^{-1}$



Determine:

a) Temperature change that occurred

(1mk)

b) Amount of ethanol used

(1mk)

c) Moles of ethanol used

(2mks)

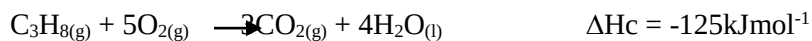
d) Amount of heat gained by water

(2mks)

e) Molar enthalpy of combustion of ethanol

(2mks)

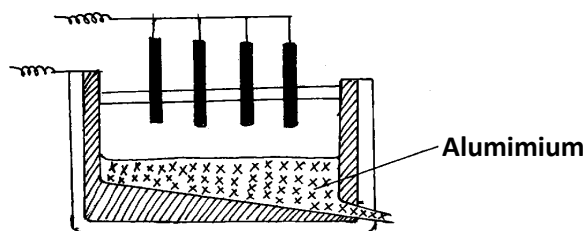
f) Use the following thermochemical processes to answer the questions that follow;



- i) Draw an energy level diagram representing the formation and combustion processes of propane, carbon and hydrogen (2mks)

- ii) Hence or otherwise, determine the heat of formation of propane (2mks)

5. The extraction of aluminium from its ore takes place in two stages, purification stage and electrolysis stage. The diagram below shows the set up for the electrolysis stage.



- (a) (i) Name the ore from which aluminium is extracted. (1mk)
- (ii) Name the impurity which is removed at the purification stage. (1mk)
- (i) Label on the diagram each of the following:-
- Anode (1mk)
- Cathode (1mk)

Region containing electrolyte. (1mk)

(ii) The melting point of aluminium oxide is 2054°C, but electrolysis is carried out between 800°C - 900°C.

I. Why is electrolysis not carried out at 2054°C. (1mk)

II What is done to lower the temperature? (1mk)

III Give **two** properties that makes aluminium be collected as shown in the diagram. (2mks)

(b) A typical electrolysis cell uses a current of 40,000 amperes. Calculate the mass in kg of aluminium produced in one hour.(3mks) IF=96500c, Al=27

6. (a) In a paper chromatography sample A was found to be more soluble than sample B. Sample C had the same solubility a sample A, while sample D was most sticky of all the samples. Mixture K contained samples B and D only. In the space provided below draw the chromatogram of A,B,C,D and mixture K (3 ½ mks)

(b) Using propanone describe how you can separate a mixture containing iodine crystals

and common salt

(2 ½ mks)

(c) Mixture of magnesium powder and zinc (II) oxide will react vigorously when heated but no reaction occurs when a mixture of magnesium oxide and zinc powder is heated

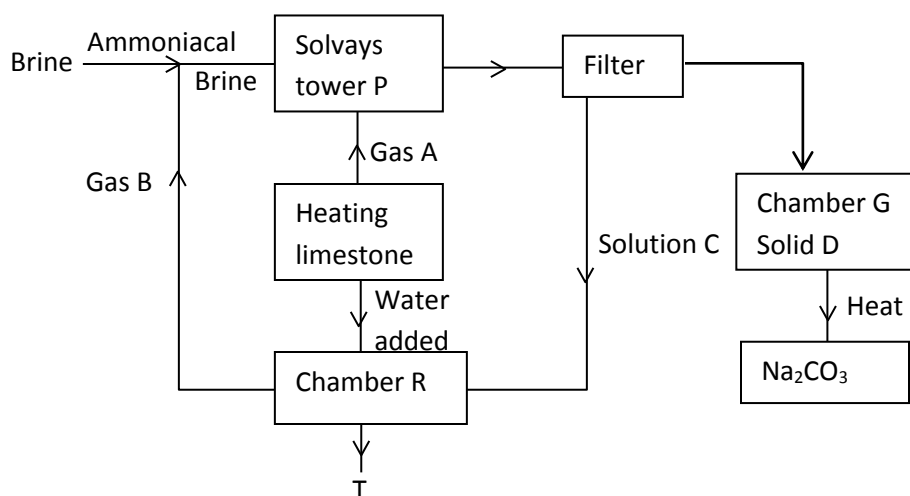
(i) Explain the observations made (1mk)

.....

(ii) Write the equation for the reaction between magnesium and zinc (II) oxide (1mk)

(iii) From the reaction above identify the oxidizing agent (1mk)

7. The diagram **below** shows the process of manufacturing sodium carbonate using ammonia soda process. Study it and answer the questions that follow.



(a) Name gases **A** and **B**. (2 marks)

A

B

- (b) Name liquid **C** and solid **D**. (2 marks)

C

D

- (c) Write equations of the reactions in: (2 marks)

Tower **P**.

Chamber **R**.

- (d) Name the product **T** formed at chamber **R** and give one of its uses. (2 marks)

- (e) Explain using ionic equations how sodium carbonate is used to soften hard water. (2 marks)

KCSE MOCKS SET 2

451/1

COMPUTER STUDIES

PAPER 1

THEORY

TIME: 2½ HOURS

SECTION A(40 MARKS)

Answer ALL questions in this section in the spaces provided.

1. State **two** practices to be observed in a computer laboratory in order to ensure the safety of the computer user
(2marks)

2. Give the name of the input device that performs the following functions:

- a) A device that converts analog signal into digital form_____ (1mark)

- b) A device that is used in Computer Aided Designing_____ (1mark)

3. Distinguish between character and line printers and give an example of each.
(2marks)

4. Give the name of the technology used by the following computer generations.
(2marks)

a) Fifth
generation_____

—

b) Second
generation_____

c) Fourth
generation_____

d) First
generation_____

—

5. a) With the aid of a diagram, distinguish between an analogue and digital signal.
(2 marks)

b) What is the equivalent of $101.100_2 + 0.0101_2$
(3marks)

6. a) Using an example of an email address, describe the main parts of an e-mail. (2
½marks)

- b) Explain what is meant by the term search engine. Give an example.
(1½ marks)

7. a) Define portability as used in software selection
(1mark)

- b) How does an operating system manage computer memory (2marks)

- c) i) What is meant by disk partitioning?
(1mark)

- ii) State the purpose of disk partitioning
(1mark)

8. Use the spreadsheet extract below to answer the following questions.

	A	B	C	D
--	---	---	---	---

1	EXAM	MATHS	ENG	CHEM
2	CAT 1	40	38	35
3	CAT 2	35	31	38
4		X	Y	Z

a) Different formulae were applied to display different results as will be indicated below.

Write the formula that would be typed at X,Y and Z to display 1400, 69 and 2 respectively

X _____

(1mark)

Y _____

(1mark)

Z _____

(1mark)

9. a) List **four** stages in data processing cycle

(2marks)

b) What is meant by the following

(3

marks)

i) Reference file

ii) Master file

iii) Report file

10. Distinguish between logical and physical computer files.
(2marks)

11. Give **two** forms of softcopy output of a TECNO H6 phone
(2marks)

12. Describe the term coaxial cables as used in data transmission and state **one** advantage of using them over twisted pair cables.
(2marks)

13. Suggest **two** ways in which fifth generation computers may be made more user-friendly for use by users with no hands.
(2marks)

14. State the keyboard shortcut for reverting an action in typing of text.
(1mark)

-
-
-
15. Define the term system control
(1mark)
-
-
-

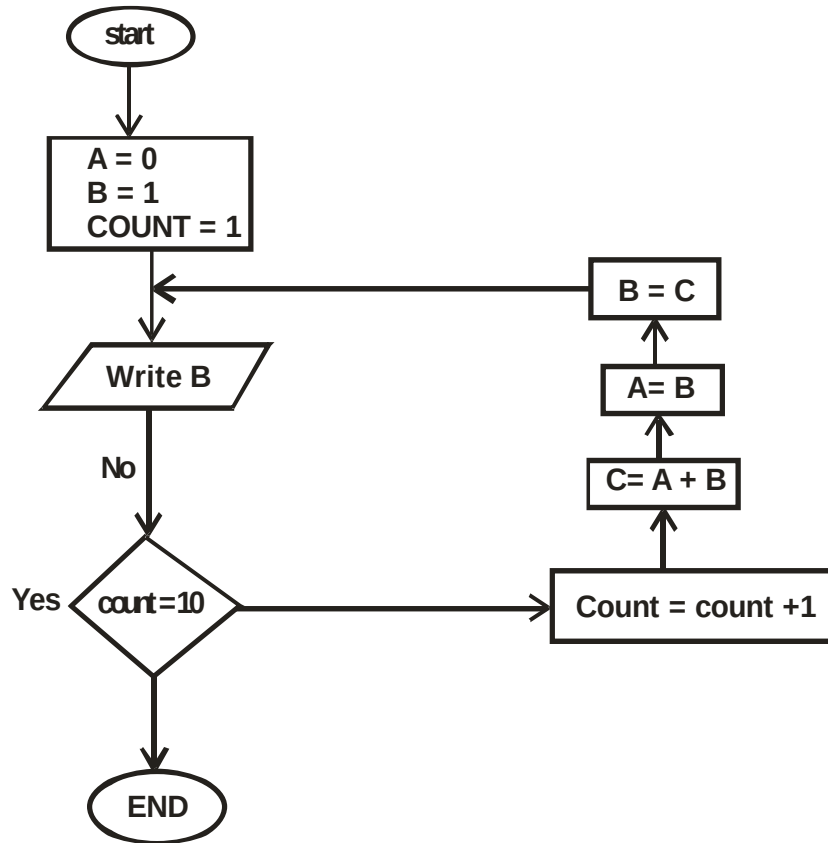
SECTION B (60 MARKS)

Answer question 16 and any other three questions from this section in the spaces provided

16. a) Identify the following programming languages
(2marks)
- i) 11111110
01011100
-

- ii) < HTML>
- <TITLE> DATA REPRESENTATION </TITLE>
- <HEAD 1>Concepts of Data Representation <HEAD >
- <BODY>Explanation of concepts
- </BODY>
- <HEAD2>Terms </HEAD 2>
- </BODY/> Nibble, byte
- </BODY>
- </HTML>
-

- b) Study the flow chart below and answer the questions that follow.



- i) Write a high level language program for the above flowchart.
(8marks)

[illegible]

- ii) List the outputs of the flow chart above.
(5marks)

17. a) Highlight **four** circumstances that can necessitate the development of a new information system by a firm.
(4marks)

- b) Mention **four** factors considered in designing a good file.
(4marks)

- c) Define the terms;

i) Attribute

(1mark)

ii) Prototype

(1mark)

d) Explain **three** characteristics of a system.

(3marks)

e) State **four** methods a system analyst uses during system design.

(2marks)

18. a) Describe how the computers have been used in reservation systems.

(3marks)

b) What is meant by the term E-commerce?

(2marks)

c) i) What does the acronym ATM stand for?

(1mark)

ii) State **two** input devices of an ATM machine.

(2marks)

d) Discuss **three** applications of artificial intelligence in expert systems.

(3marks)

e) Mention **four** reasons why employees may resist introduction of ICT to their place of work.

(4marks)

19. a) Which key on the keyboard can play the role of the start button?
(1mark)

- b) Give **two** uses of the task manager.
(2marks)

- c) One characteristics of an operating system is to control computer. State **two** resources under the operating system control.
(2marks)

- d) Explain **five** features of a word processor.
(5marks)

- e) Explain the following concepts as used in spreadsheets.
(4marks)

- i) Automatic recalculation

- ii) ‘what if’ analysis

f) State the least component of a database hierarchy.
(1mark)

20. a) Subtract 0111_2 from 1001_2
(2marks)

b) Using two's complement, subtract 7 from 4 and give the answer in decimal number system.
(4marks)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

c) Convert;

i) $91B_{16}$ to binary notation
(3marks)

ii) 376_8 to decimal
(3marks)

iii) 9.625_{10} to binary
(3marks)

KCSE MOCKS SET 2

451/2

COMPUTER STUDIES

PAPER 2

(PRACTICAL)

TIME: 2½ HOURS

1. a) Kibos holding company limited has given you the details below to be used in calculating expenses incurred during two quarters of the year as indicated. Use a spreadsheet package to enter the details, and save your work as **SAMBU 1**.

(10 Marks)

	A	B	C	D	E	F	G
1	<u>KIBOS HOLDINGS COMPANY LIMITED</u>						
2	<u>P.O BOX 23625 SAMBU</u>						
3		First quarter			Second quarter		
4		January	February	March	April	May	June
5	Production Cost	15642	14687	18741	19457	15412	15441
6	Transportation	1564	1469	1874	1946	1541	1544
7	Warehousing	1125	1056	1347	1398	1107	1109
8	Promotion	2564	2407	3071	3188	2525	2530
9	Salary	4525	4248	5420	5626	4456	4465

- b) i) Copy the details of Kibos holding company limited to sheet 2 of your worksheet.

(2marks)

- ii) Add a column called “Total” and calculate the Total production cost of the two Quarters.

(4marks)

- iii) Use relative referencing to calculate the total for the other expenses. (4marks)

c) i) Calculate the total expenses for each month so as to give the results as “Total expenses” on row 10.

(4marks)

ii) On the paper provided, write the formula used to calculate the expenses for the month of April.

(2marks)

d) i) Use absolute referencing to calculate the cumulative total expenses for the Two quarters in cell H10.

(4marks)

ii) Write the formula used to get the cumulative total on the paper provided. (2marks)

iii) Calculate the percentage of total expenses of the respective months and display the results on row 11 as “percentage of total expenses”.

(4marks)

iv) Write the formulae used to get the percentage of May. (2marks)

v) Draw a line graph to illustrate the percentage of the total expenses for the Six months indicating the months. Save your work as SAMBU 2.

(10marks)

vi) Print SAMBU 1 and SAMBU 2 (2marks)

2 a) A hospital uses a database to maintain data about its employees. Create a database file named **EMPLOYEE**.

(1mark)

b) Create a table called **EMPLOYEE 1** with the following fields and hence enter data into it as shown in fig 1 below. NB: Choose an appropriate primary key.

(14marks)

- EMP NO
- NAME
- DATE OF BIRTH
- DEPARTMENT
- BASIC PAY

Fig 1

Emp	Name	Date of birth	Department	Gross pay
-----	------	---------------	------------	-----------

01	John Kirui	16/2/1972	Computer	28000
02	Margaret Wairimu	2/2/1960	Medical	30000
03	Jane Cherono	2/5/1970	Management	15000
04	Victor Oduor	8/12/1981	Accounts	25000
05	Harry Wanyama	23/6/1983	Medical	30000
06	Jacob Kiprono	19/4/1973	Management	45000
07	Ali Mohammed	1/1/1969	Medical	30000
08	Daniel Omondi	3/5/1983	Accounts	25000
09	EverlyneKitune	11/3/1971	Medical	20000
10	Nancy Kerubo	22/9/1980	Medical	20000

c) i) Insert Two new fields to hold the employee's Profession and deductions.

(4marks)

ii) Data for included fields is as follows;

Profession

Mohammed, Wanyama and Wairimu are Doctors. Oduor and Omondi are accountants. Kiprono is an Administrator, Kirui is a systems analyst and Cherono is a secretary.

Deductions

Emp no	Amount
07, 05, 02	8,000
04, 08	7,500
06	10,000
01	12,000
03	3,500
09, 10	4,000

Enter the above data into the respective fields.

(6marks)

iii) Sort the records in ascending order based on the name field and save as **EMPLOYEE**

2.

(3marks)

d) i) From **EMPLOYEE 2**, extract a list of employees who were born between 1960 and 1972 both years inclusive and are accountants, nurses or doctors. Write down on paper provided the query expression you used to extract the data. Save the extracted list as **LIST1**.

(8marks)

ii) Remove the date and occupation criteria on **LIST 1**. Add a calculated field to calculate the Net pay of all employees as Gross Pay – Deductions. Save as **LIST**

2. (5marks)

e) Generate a columnar report based on **LIST 2** with the following fields; Emp no, Name, Department and Net Pay. Sum up the Net Pay for all employees in the report. Save your report as **REPORT**.

(6marks)

f) Print **EMPLOYEE1, EMPLOYEE2, LIST1, LIST 2** and **REPORT**.

(3marks)

KCSE MOCKS SET 2

101/1

ENGLISH
Paper 1
Functional skills

2 Hours

1. Your drama club is organizing to stage a performance of the play, Caucasian Chalk Circle. You are inviting neighbouring schools and the general public to the event.

- a. Write the notice announcing the event and inviting other schools and the public to attend.(8mks)
- b. In not more than 200 words, write a synopsis of the play to accompany the notice (12mks)

2 Fill in the blank spaces with the most appropriate word.

As you think about the different aspects that go.....1.....your persona, it is important to.....2.....your best traits and recognize the.....3.....you may have been taking for granted all.....4..... . Doing this will provide a foundation of confidence that serves as a.....5.....to connect you to other people.

.....6.....your appearance is what makes people curious to get to know you, your presence is what makes them respond to you. Your presence is your energy, your spirit and your aura. It includes.....7.....those qualities that help you bond with another human being. It starts with eye contact and moves into your listening skills as well as the extent to which you are engaged in a conversation. Some people are.....8.....there, giving 100 percent; others give 20 percent at.....9.....because their minds are elsewhere.

Eye contact is an.....10.....powerful tool. So long as you are making eye contact while talking, the topic is not likely to matter much.

3 a. Against each of the following sentences, indicate whether you would end with a rising or falling intonation. (5mks)

1. You actually saw the pyramids?.....
2. My younger sister has identical twins.
3. Why did you oversleep?.....
4. Could I come with you, please?.....
5. What a tragic experience that was!.....

b. *Read the following poem and then answer the questions*

Lazy bones, let's go to the farm
Sorry, I've got a headache
Lazybones, let's go pounding grains
Sorry, my leg isn't right

Lazybones, let's go fetch firewood
 Sorry, my hands are hurting
Lazybones, come and have some food
 Hold on, let me wash my hands!

i. How would you say Lazybones' responses in line 2, 4 and 6? Give reasons for your answer. (3mks)

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ii. If you were performing this poem, how would you say the last line? (2mks)

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iii. Provide another word that has the same pronunciation as each of the following words (3mks)

some.....

Right.....

Come.....

C. Your school choir is rehearsing a choral verse for the school's music festivals. They are making a presentation for the rest of the school so as to get some feedback. Give **three** things that the listeners need to pay attention to and explain why. (3mks)

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D. You are part of an audience that is listening to a speech. You look around and notice that some people are looking at their watches, a few are yawning, and two are shifting in their seats. State three things that you would do to ensure you continue listening effectively (3mks)

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E. Read the dialogue below and then answer the question that follows

Buyer: How are you this morning?

Seller: I'm okay

Buyer: I'm looking for good chicken, but yours don't look too good.

I'm going to have visitors, and this being Christmas season, I really must give them a feast

Seller: These are the right kind of chicken for your visitors. They're healthy and well fed.

Buyer: On the contrary, they look underfed. Anyway, what is your price?

Seller: It depends I charge more for cocks; they have more meat, you know (pointing at a red cock).

This one, for instance, goes for ksh 400.as for hens, I charge ksh 250

Buyer: You're not serious! Much of the weight is a bundle of bones. I 'm giving you ksh 150 for each hen and ksh 300 for each cock. I'm buying three of each-three hens for ksh 450 and three cocks for. sh 900.this will give us a total of ksh 1350.

Seller: You know, I buy and sell. I don't get them from my shamba. Your figure doesn't give me any profit at all.

Buyer: But you also know money is hard to come by, and especially during this Christmas season. Give me a reasonable price, unless you prefer I go to another seller.

Seller: Let me make it sh 225 for a hen and sh 375 for a cock.

Buyer: it looks like you're not interested in selling your chickens.

Seller: No, I'm, why would I be here? My children's fees come from this business.

Buyer: Okey, take sh 175 for each hen and sh 325 for each cock.

Seller: No, there would be no profit for me. You can do better than that. You can surely promote my small business. Give me sh 350 per cock and sh 225 per hen. This would be sh 625 for the three hens and sh 1050 for the three cocks.

Buyer: (Doing mental arithmetic) That's a total of sh 1725. Okay, at least I'll be able to feed my visitors. (handing over the money) hare you are.

Seller: Thank you.(as the seller ties them together) You're good customer. Please come again, my name is Musimbi

Buyer: And I'm Karani, see you then.

Seller: See you

I What is the purpose of the greetings in this situation (1mk)

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.....

ii. Identify and explain the negotiation skills of the buyer (5mks)

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F. You have been invited for a very important interview.

a) Explain briefly three things you would need to do before the occasion to ensure that you **look** presentable.(3mks)

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b) What else would you need to do **during** the interview to ensure your success (2mks)

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.....
.....

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CALL/TEXT/WHATSAPP 0705525657
FOR ENGLISH PAPER 2&3
QUESTIONS

KCSE MOCKS SET 2

Kenya Certificate of Secondary Education

GEOGRAPHY

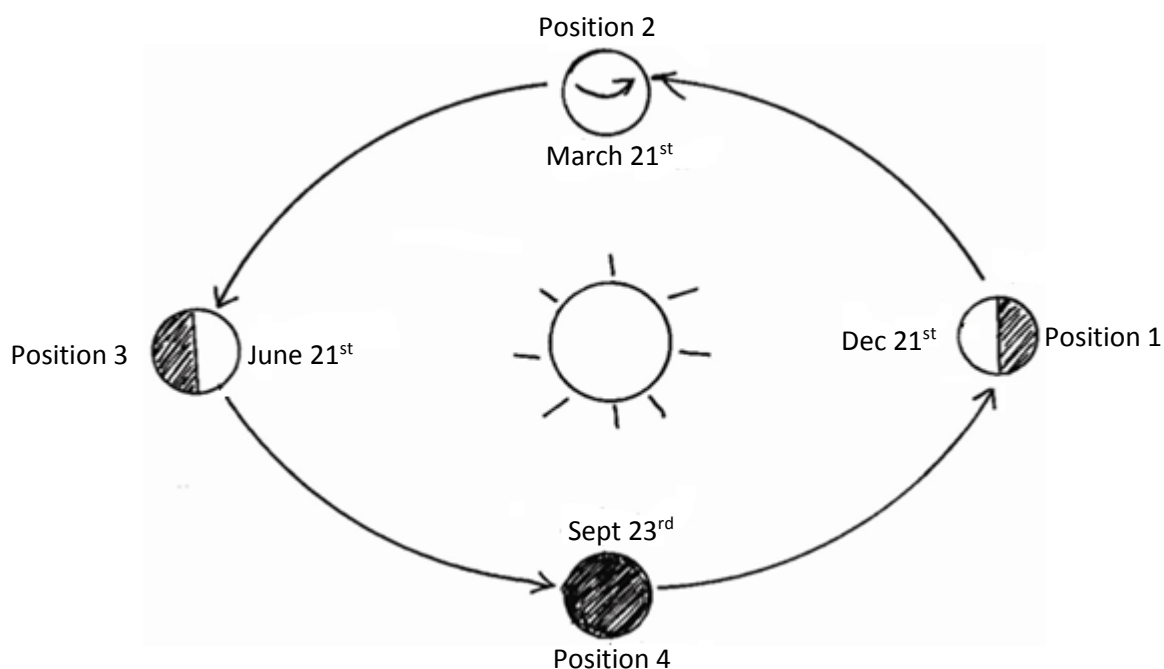
PAPER 1

TIME: 2¾ HOURS

SECTION A:

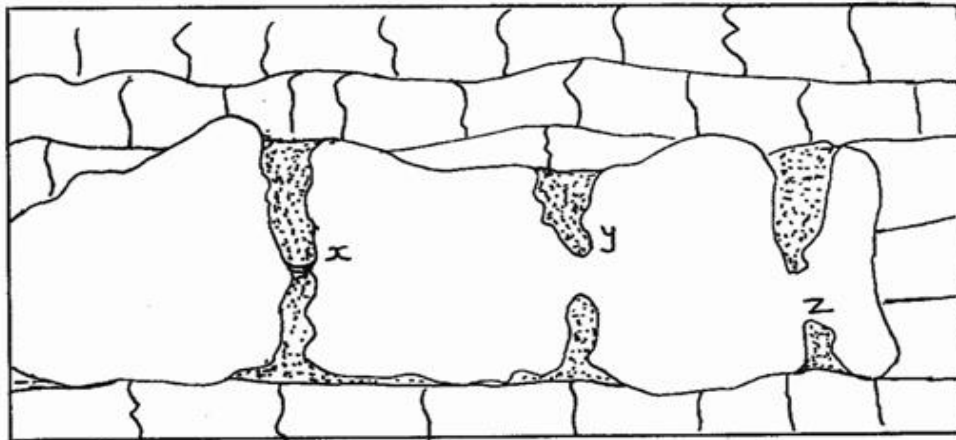
Answer **all** questions from this section in the spaces provided.

1. (a) Use the diagram **below** to answer question a(i) and (ii)



- (i) What season does the Northern Hemisphere experience at position I? (1mk)
- (ii) Which **two** positions does the earth experience equinoxes (2mks)

- (b) State **two** characteristics of the lower mantle. (2mks)
2. (a) Name **two** types of seismic waves. (2mks)
- (b) State **three** major earthquake zones of the world. (3mks)
3. (a) Differentiate between weathering and mass wasting. (2mks)
- (b) Apart from living organisms give three other factors that influence the rate of weathering. (3mks)
4. (a) Define aridity. (2mks)
- (b) State reasons why wind is the most effective agent of erosion in arid areas (2mks)
5. The diagram below represents underground features in a limestone area.
Use it to answer question (a).



Geography Paper 1

2

- (a) Name the features marked X, Y and Z. (3mks)
- (b) State **three** factors that influence the development of karst scenery (3mks)

SECTION B:

Answer question **6** and any other **two** questions from this section.

6. Study the map of Kitale 1: 50,000 (sheet 75/3) provided and answer the following questions.
- a) i) Name **two** scales used in the map. (2mks)
- ii) Identify **two** natural features found in the grid square 4711. (2mks)
- iii) What is the altitude of the highest point in the area covered by the map. (2mks)
- b) i) What is the bearing of the tank at grid square 2726 from the trigonometrical station at grid square 2823. (2mks)
- ii) Measure the distance of dry weather road (C 640) from the junction at point **M** (345142) to the junction at point **N** (416201). Give your answer in Kilometres. (2mks)
- c) i) Calculate the area of Kitale municipality. Give your answer in sqkm (2mks)
- ii) Draw a rectangle (6cm by 8cm); a reduced map to represent the area bounded by eastings 33 and 39 and northings 11 and 19. On the map mark and name:
- An estate
 - road
 - A bridge
 - A camp
- d) i) Citing evidence from the map, identify **four** economic activities in the area covered by the map. (4mks)
- e) Identify **three** vegetation types in the area covered by the map. (3mks)
7. (a) (i) Differentiate between weather and climate. (2mks)
- (ii) Identify any **three** elements of weather. (3mks)

(b) State the apparatus found in a Stevenson Screen. (3mks)

(c) Describe the factors that influence temperature. (5mks)

(d) The table below shows rainfall and temperature figures of a station in Africa. Use it to answer questions d(i), (ii).

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp. °C	23	24	23	22	19	17	18	19	19	20	22	23
Rainfall/mm	109	122	130	76	52	34	28	38	70	108	121	120

(i) Calculate annual range of temperature (show your calculations). (2mks)

(ii) Calculate the annual rainfall. (2mks)

(e) (i) What is weather forecasting? (2mks)

(ii) Explain the significance of weather forecasting to human activities. (6mks)

8. (a) (i) What is earth movement? (2mks)

(ii) Identify **two** types of earth movements. (2mks)

(iii) State **three** causes of earth movement. (3mks)

(b) Describe the continental drift theory in the formation of continents (5mks)

(c) With aid of diagram describe how the Rift Valley was formed by compression forces. (6mks)

- (d) Explain negative effects of faulting and human activities. (8mks)

9. a) i) What is a Karst scenery?

(3mks)

- ii) Name **two** features found in each of the following parts of a Karst scenery.

(2mks)

- Surface - Underground

- b) Explain **three** factors which influence the formation of features in the Karst scenery.
(6mks)

- c) State the significance of a Karst scenery.
(4mks)

- d) You are supposed to carry out a field study of an area eroded by water.

- i) Give **three** reasons why you would need to prepare a working schedule.
(3mks)

- ii) Why do you need a map of the area of study.
(2mks)

- iii) State **three** recommendations that you would make from your study to assist the local community to rehabilitate the eroded area.
(3mks)

10. (a) Define the term hydrological cycle
(2mks)

- (b) Using a well labelled diagram describe the processes through which circulation of water is carried out. (8mks)

- (c) Explain the significance of the hydrological cycle.
(8mks)

- (d) Students from Gawa Secondary are planning to carry out a field study on sources and uses of water around their school.
- (i) Identify any **three** activities they would carry out before the actual field study.
(3mks)
- (ii) Give **four** reasons why it is important to have a working schedule.
(4mks)
- (iii) State any **three** methods they would use to record data. (3mks)

KCSE MOCKS SET 2

312/2

GEOGRAPHY PAPER 2

TIME: 2 ¾ HOURS

SECTION A

Answer all the questions

1. (a) What is arable farming (2mks)
(b) Give three physical condition favouring growth of sugar cane in Kenya (3mks)
2. (a) State two factors affecting distribution of forest in Kenya (2mks)
(b) State three advantages of forests (3mks)
3. (a) Give the difference between Immigration and Emigration (2mks)
(b) State three factors leading to migration (3mks)
4. (a) What is Eco-Tourism (2mks)
(b) Give three differences between tourism in Kenya and Switzerland (3mks)
5. (a) Apart from wind name two environmental hazards (2mks)
(b) Write down three effects of wind on the environment (3mks)

SECTION B

6. (a) (i) Use the data below to draw comparative line graph. (7mks)

Vehicles	1990	1991	1992	1993
Car	16	18	16	11
Buses	29	29	27	19
Lorries	40	39	37	30
Vans	7	8	7	14

- (ii) Give three advantages of the Graph you have used (3mks)
- (b) State three advantages of using road transport over air transport (3mks)

- (c) Explain three physical conditions which hinders the development of river transport in Kenya (6mks)
- (d) (i) State two forms of communication used in Kenya today (2mks)
(ii) Give four advantages of containerization (4mks)
- 7. (a) Differentiate between fishing and fisheries (2mks)
(b) State three conditions favouring development of planktons (3mks)
(c) Explain four factors favouring fishing in Japan (8mks)
(d) Give three reasons why the government of Kenya is encouraging fish farming (3mks)
(e) (i) Explain three problems facing marine fishing in Kenya (6mks)
(ii) State three possible solutions for the above problems in e (3mks)
- 8. (a) Define the term industrialization (2mks)
(b) Explain four factors that led to the development of iron and steel industry in Ruhr region of German (8mks)
(c) State four characteristics of cottage industry (4mks)
(d) Give four ways in which the Kenyan Government encouraging Jua Kali industry (4mks)
(e) Form four students are intending to carry out a field study on Jua Kali industry
(i) State two objectives of their study (2mks)
(ii) Identify two possible problems faced by Jua Kali artisans (2mks)
(iii) State three follow-up activities of the study (3mks)
- 9. (a) Name two types of trade (2mks)
(b) Explain five problems facing trade in East Africa (10mks)
(c) State four benefits that Kenya derives from international trade (4mks)
(d) Explain three factors influencing the future of trade in Kenya (6mks)
(e) Name three products that are imported to Kenya (3mks)
- 10. (a) Give the difference between beef farming and pastoralism (2mks)

- (b) (i) State four characteristics of pastoralism (2mks)
- (ii) Explain three problems facing pastoralism in Kenya (6mks)
- (c) i) State four physical conditions favouring beef farming in Argentina (4mks)
- ii) Compare beef farming in Kenya and Argentina (6mks)
- iii) State three significance of Beef farming (3mks)

KCSE MOCKS SET 2

HISTORY 311/1

NAME.....ADM NO

The paper consist of three sections A, B &C

Answer all questions in section A, three questions in section B and two questions in section C

SECTION A. 25 MARKS. Answer all questions in this section

1. Give the main source of information on pre-history. 1 mk
2. Name two dispersal areas of the coastal Bantus. 2mks
3. Give two political roles of the Orkoiyot. 2mks
4. Identify two types of written evidence that support the presence of early visitors. 2 mks
5. Name one political responsibility of a Kenya citizen. 1 mk
8. Give two social factors that promote national unity. 2 mks
7. Identify the type of constitution used in Kenya. 1 mk
8. Give two special rights of the older members of the society. 2mks
9. Two communities that shared mixed reaction to the Europeans. 2 mks
- 10 Identify two roles of Chief and Headmen in colonial Kenya. 2 mks
12. Give the main reason why Africans were not allowed to grow cash crops. 1 mk
13. Identify one major reform which resulted from the Lyttelton constitution of 1954 in Kenya?(1mk)
14. Identify two methods that the colonial government used to discourage the activities of the Mau Mau movement in Kenya.(2mks)
15. Give the main contribution of Professor Wangari Maathai. 1 mk
16. Who swears in the speaker of the National Assembly after being elected? 1 mk

17. Identify the latest railway development in Kenya. 1 mk

SECTION B (45mks)

Answer any three questions from this section in the answer booklet provided.

18a) Give five reasons for the migration of the Luo from their original homeland into Kenya. (5mks)

b) Describe the political organization of the Agikuyu during the pre-colonial period (10mks)

19a) Give five reasons why the early visitors from Arabia came to the Kenya coast before 1500. (5mks)

b) Explain five factors which contributed to the Portuguese success in their conquest of East Africa Coast (10mks)

20a) Give three characteristics of Independent Churches movement during the colonial period. 3 mks

b) Explain six factors that hindered the formation and operation of the early political movement in Kenya. 12 mks

21a) Outline the demands of the white settlers contained in the Devonshire white paper of 1923 (3mks)

b) Explain six problems encountered by white settlers in Kenya. 12 mks

SECTION C. (30MKS). ANSWER TWO QUESTIONS ONLY

22.a) State five reasons why national integration is important in Kenya. 5 mks

b) Explain five methods of conflict resolution. 10 mks

23a) Identify three factors that are likely to interfere with free and fair elections in Kenya. 3 mks

b) Explain six functions of the Independent Electoral and Boundaries commission of Kenya. 12 mks

24a) State three characteristics of African socialism. (3mks)

b) Discuss the impact of the Harambee philosophy in Kenya's development (12mks)

KCSE MOCKS SET 2

HISTORY AND GOVERNMENT

311/2

THIS PAPER HAS THREE SECTIONS, A, B & C.

SECTION A, 25 MARKS. ANSWER ALL QUESTIONS IN THIS SECTION

1. Define the term history. 1 mk
2. Give two distinct practices developed by Homo-Erectus. 2mks
- 3 Identify two methods of irrigation used by early Egyptians. 2mks
3. State the main characteristic of the Stone Age era. 1 mk
5. Name two main exports from West Africa during Trans – Saharan Trade. 2mks
6. Mention the main sources of energy during the early Industrial Revolution. 1mk
7. Give two factors that contributed to Agrarian Revolution in Britain. 2mks
8. Give two transport systems that encouraged industrial growth in Britain in the 19th century. 2mks
9. What was the importance of the Golden stool in Asante's kingdom in the 19th century? 1mk
10. State two functions of Lukiko in the Buganda kingdom in the 19th century. 2mks
11. Why is Berlin City in Germany important in Africa colonial history? 1mks
12. Why did the British apply indirect rule in Nigeria. 1mk
13. Name the party that led Ghana to independence. 1mk
14. What's the importance of the year 1994 in South Africa history? 1mk
15. State the immediate cause of the First World War. 1mk
16. Give two reasons why Russia withdrew from World War I in 1917. 2 mks
17. Name one UN agency based in Kenya. 1 mk

SECTION B

ANSWER ANY THREE QUESTIONS FROM THIS SECTION

- 18.(a) State three theories of the origin of man. 3mks
(b) Explain the way of life of the early Man in the upper Paleolithic period. 12mks
19. (a) Name three indigenous crops which Africana should grow to reduce food shortages. 3mks
(b) Explain six causes of food shortages in Africa. 12 mks
20. (a) Give five roles performed by the Tuaregs in the Trans – Saharan Trade. 5mks
(b) Explain the result of Trans- Saharan Trade. 10 mks
21. (a) What were the uses of gold in Egypt. 5mks
(b) Describe the effects of iron-working in Africa. 10mks

SECTION C

ANSWER ANY TWO QUESTIONS FROM THIS SECTION

22. (a) Give five reasons why Lewanika collaborated with the Europeans. 5mks
(b) Explain the outcome of Lewanika's collaboration with Europeans. 10mks
23. (a) Name three European nations that scrambled for West Africa in the 19th century. 3mks
(b) Explain six reasons why Samori Touré was able to resist the French for a long time. 12 mks
24. (a) Give 3 reasons why U.S.A. joined the First World War. 3 mks
(b) Explain reasons why the Allied Powers won the First World War. 12 mks

KCSE MOCKS SET 2

441/ 1

HOME SCIENCE (THEORY)

PAPER 1

2 ½ HOURS

SECTION A – 40MKS (COMPULSORY)

Answer all the questions in this section

1. Give **two** functions of calcium in the body. (2mks)

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2. Suggest **four** uses of eggs in flour mixtures. (2mks)

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3. Give **two** reasons why common salt is iodized. (2mks)

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4. Green vegetables should be blanched before freezing. Give **three** reasons why. (2mks)

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5. Give **two** functions of pleats. (2mks)

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6. State **two** disadvantages of using sodium bicarbonate as a raising agent. (2mks)

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7. Identify **two** methods used in the repair of clothes and household articles. (2mks)

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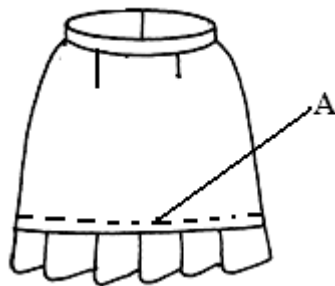
8. Give **two** disadvantages wool has over cotton. (2mks)

.....

.....

.....

9. a) Identify the stitches (A) used to attach the fill onto the skirt below. (1mk)



b) Give other 2 areas where the stitch can be used. (2mks)

10. Write down **three** factors to consider when selecting fabric for baby clothes. (2mks)

.....

.....

.....

11. State three negative effects of advertisement on the consumer. (3mks)

.....

.....

.....

12. State two dangers of smoking during pregnancy. (2mks)

.....
.....
.....
13. State two functions of sodium in the body. (2mks)

.....
.....
.....
14. Classify permanent stitches. (3mks)

.....
.....
.....
15. State three attributes of a well groomed person. (3mks)

.....
.....
.....
16. Give three reasons for hanging clothes on cloth line? (3mks)

.....
.....
.....
17. Name three types of stitches used for neatening edges? (3mks)

SECTION B (20 MARKS) COMPULSORY

18. Your aunt from Mlolongo has visited you over the weekend. Describe how you would clean:

(a) the painted wall of the guest room. (8mks)

(b) A polished wooden stool. (8mks)

(c) clean an enamel bowl. (4mks)

SECTION C (40 MARKS)

Answer any TWO questions from this section in the spaces provided after question 22

19. (i) Explain two reasons for repairing garments. (4mks)
(ii) Give six points to bear in mind when mending garments. (6mks)
(iii) With the aid of diagrams, describe how to mend a worn out plain coloured bed sheet. (12mks)
20. a) Give **three** functions of teeth. (3mks)
b) State **three** factors to consider when choosing laundry basin. (3mks)
c) Explain **three** advantages of comparative shopping. (6mks)
d) Describe **four** points on conservation of energy in lighting. (8mks)
21. (a) State three advantages of lining curtains. (3mks)
(b) Outline six precautions to take when using electrical appliances in the home. (6mks)
(c) Identify five activities carried out in management of amoebic dysentery. (5 mks)
(d) Highlight six points to observe when looking after a sick person. (6 mks)

KCSE MOCKS SET 2

102/1

KISWAHILI

Karatasiya kwanza

Insha

MUDA:SAA 1 $\frac{3}{4}$

SWALI LA 1 (LAZIMA)

1. Umeteuliwa kuwa miongoni mwa watakaohutubia wananchi kuhusu hali ya usalama nchini Kenya. Andika hotuba utakayowasilisha siku hiyo kuhusu visa vya ukosefu wa usalama na upendekeze hatua zinazofaa kuchukuliwa ili kukabiliana na hali hii.
2. Tetea kauli kuwa Dunia ni 'kijiji tandawazi'.
3. Andika insha itakayodhirisha kuwa "Afadhali kujikwaa kidole kuliko ulimi".
4. Andika insha itakayomalizika hivi:-
..... Jioni hiyo chajio kilinishinda, nikawazia hotuba ya Waziri wa Usalama kuhusu visa vya kudorora kwa usalama na mauaji yaliyokithiri.

KCSE MOCKS SET 2

KISWAHILI

Karatasi ya 2

(LUGHA)

Muda: Saa: 2½

1. UFAHAMU (ALAMA 15)

Soma taarifa ifuatayo. Kisha ujibu maswali yanayofuata.

Kwa kipindi cha miezi kadha iliyopita tumeshuhudia vyombo vya dola vikitia makali yake kwenye upekuzi na hata kupiga doria usiku na mchana katika jitihada za kulinda nchi.

Mpango huu ni kweli umeonekana kufanya kazi hivi kwamba hata magaidi wenyewe wameshindwa kupenya katika miji na sasa kuhiari kijinga kushambulia magari ya abiria kitendo ambacho ni cha kuonyesha uwoga. Kwa hatua hiyo, navipa vyombo vya dola kongole. Jambo ambalo lafaa kujulikana ni kwamba mikakati ya kulinda nchi haifai kuwa ni ya wakati mmoja tu, mbali inafaa kuwa ni zoezi la kila siku.

Magaidi nao huwa macho huku yakijua bayana kwamba wakati wa kulala kwa walinzi unapokuwepo, basi wanapata nafasi ya kututupia ‘viazi’ ukipenda grunedi.

Kama ilivyo kawaida katika mataifa mengi barani Afrika, ni bayana kwamba bado kungali na mianya mingi ambayo magaidi wa kimataifa huendelea kutumia. Dosari bado zipo. Kwa mfano mipaka mingi ya nchi hizi huwa kama lango kuu la ugaidi wa kimataifa, kwani kuenea kwa saratani ya ufisadi halimo tu maofisini mbali pia kwenye mipaka yetu.

Kama kupata kitambulisho, pasipoti na stakabadhi zingine za kusafiri nchini Kenya ashakum si matusi ziligeuzwa ‘maandazi’ ya kariakoo basi niambie ni nani hawezi kuingia na kutoka nchini bila usumbufu wowote ule bora tu anayehitaji ana hela mkononi? Kwa kuikubali hongo kuwa ufunguo wa kila kitu, Wakenya wenzangu hapo naona ni kama tumejiweka kwenye kikaango kilicho juu ya moto mkali. Hapa hakuna aliye na bahati, tajiri kwa masikini wamo kwenye mtego huu hatari.

Kwa mtindo ambao tunafuata wa kutoa ajira katika idara mbalimbali za ulinzi, inabidi serikali iwe na uangalifu sana hasa kwenye suala nzima la kuhakikisha stakabadhi wanazohitaji si ghushi.

Pasina kufanya hivyo hapo tena tunaweza kuwapata maadui wanaopenya na kujifanya walinzi wetu kumbe ni majasusi wa magaidi. Kila Mkenya anafaa kujihisi kulindwa. Miji, vijiji na hata vitongoji vinafaa kuwa na usalama wa kutosha, kwani kila Mkenya ni mlipa ushuru na hatufai kuona labda tabaka la juu likipendelewa huku mitaa ya mabanda ikiachiwa mbwa koko kama walinzi wao.

Suala lingine muhimu ni kuangaziwa upya usalama kwenye magari ya usafiri.

Juzi tulishuhudia mabasi mawili yakilipuliwa kwenye barabara ya Thika huku tukijua fika kwamba, mpango walinda nchi ungalipo.

La kusitikitisha ni kuona kwamba, madereva na utingo wao walikamatwa na kufunguliwa mashtaka kwa kutozuia shambulizo hilo. Je, hii ni sheria gani? Dereva ataendesha gari au atachukua jukumu la walinda usalama?

Waswahili walinena kwamba mkubwa nijaa na kwa hivyo Rais wa taifa ndiye anayefaa kubeba mzigo mzima wa usalama wetu bila kubananga wasaa. Wengine ambao wanafaa kuwajibika ni wakuu wote wa idara mbalimbali za usalama.

Usalama wako na wangu ni muhimu, elewa bayana kwamba bila usalama watalii hawawezi kuja kututembelea. Bila ya usalama maendeleo ya taifa kamwe hayawezi kupatikana, ndiposa kila jitihada sharti zifanywe ili wote waweze kuendelea kuyafurahia matunda ya uhuru wetu.

Ada ya mja kunena muungwana ni vitendo, hatufai tu kuimba wimbo wa Linda nchi, ilhali mabasi barabarani hayana usalama. Hebu na tuuone ulinzi endelevu na hapo tutawakomoa magaidi kwa yakini.

Maswali

1.Ipe taarifa hii anwani mwafaka.

(alama 1)

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2.Kwa nini magaidi wanashambulia magari ya abiria?

(alama 2)

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.....

3.Eleza njia tatu ambazo zinatumiwa na magaidi kutekeleza unyama wao.

(alama 3)

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.....

.....

4.Ni njia gani ambazo magadii hutumia kuingia nchi wanazoazimia kutekeleza uhalifu? (alama2)

.....

.....

.....

5. Kwa nini suala la usalama kwenye magari ya usairi ni muhimu?

(alama 1)

.....
.....
6.Ni njia zipi zinazoweza kutumiwa kupunguza mashambulizi ya kigaidi kulingana

na mwandishi?

(alama 2)

.....
.....
.....
7.Eleza athari za utovu wa usalama.

(alama 2)

.....
.....
8. Eleza neno au mafungu ya maneno kama yalivyotumiwa katika kifungu. (alama 2)

(a)Vyombo vya dola

.....
(b)Jaa

.....
UFUPISHO (ALAMA 15)

Soma kifungu kifauatacho kisha ujibu maswali.

Mifuko ni muhimu sana katika maisha ya binadamu ya kila siku. Baadhi ya mifuko huundwa kwa namna ambayo inadumu na inaweza kutumiwa mara kadha wa kadha. Hii huweza kuhifadhiwa na kutumiwa kwa kipindi kirefu. Asilimia kubwa ya mifuko ya aina hii hutengenezwa kutokana na nguo au ngozi. Hata hivyo, ipo mifuko mingine ambayo si ya uashi, haidumu. Hii ni mifuko myepesi, rahisi kubebeka na inayopatikana kwa wingi sana. Hii ni mifuko ya plastiki.

Mifuko ya plastiki hupatikana katika maeneo mengi sana. Mifuko hii sio ghali inapolinganishwa na ile ya nguo au ya ngozi. Hii ni ya gharama ya chini na hupendwa kutokana na wepesi wake. Hata hivyo, mifuko hii inaweza kuwa chanzo cha matatizo mengi. Matatizo haya hutokana na linalopasa kufanywa kuhusu mifuko hii baada ya kutumiwa kwake. Mifuko ya nguo ambayo inatupwa kwenye jalala huishia kuoza na kuwa sehemu ya uchafu wa jalala hilo. Kwa upande wake, mifuko huo. Hata inapofukiwa arthini haiwezi kuoza hata kama kufukiwa huko ni kwa miaka mingi. Aidha hata pale inapochomwa, haiwezi kuteketea hadi kuwa jivu kama ilivyomifuko ya karatasi.

Upo umuhimu mkubwa wa kuwepo kwa sera madhaubuti za kupambana na tatizo la mifuko hii. lazima zitungwe sheria ambazo zinakabiliana na uchafuzi wa mazingira kwa njia hii. kwa njia hii hatari zianzowakabili watu na wanyama zitapunguzwa kwa kiasi kikubwa. Kwa hakika njia mojawapo ambayo ingeweza kutumiwa ni kuharamisha matumizi ya mifuko hii katika upakiaji. Sheria zinaweza kupitishwa ambazo zinahimiza matumizi ya mifuko mbadala kama ile ya karatasi kubebea vitu vidogo Fauka ya hayo, pana haja ya kuwazia kuwepo kwa njia nzuri kuteketezea mifuko hiyo. Inawezekana kuhimiza watumiaji kuirejesha mifuko hiyo mahali Fulani kwa ajili ya uteketezaji huo. Hata hivyo, hali hii huenda ikawa ngumu kwa kuwa inahitaji nidhamu ya hali ya juu sana kwa upand ewa watumiaji.

(a)Fupisha ujumbe wa aya tatu za mwanzo kwa maneno 105 – 110. (alama 10, 1 utiririko)

Matayarisho

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Nakala safi

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(b) Eleza mawazo makuu yanayojitokeza kaitika aya ya mwisho (maneno 50 – 55) (alama 5, 1 utiririko)

Matayarisho

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.....

Nakala safi

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MATUMIZI YA LUGHA (ALAMA 40)

(a) Tunga sentensi kubainisha muundo ufuatao.

(alama 2)

KN(W+V)+KT(T+E)

.....

.....(b) Sauti /ch/ ina sifa

zipi?

(alama 2)

.....

.....

(c) Andika kwa ukubwa wingi.

(alama 2)

Mwizi aliiba nguo na kiti

.....

(d)(i) Tofautisha shamirisho kipozi na shamirisho kitondo.

(alama 1)

.....

.....

.....

.....

ii) Tunga sentensi moja kuonyes shamirisho kipozi na kitondo.

(alama 2)

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.....

.....

(e) Andika upya sentensi zifuatazo kwa kuzingatia kauli zilizomo mabanoni.

(alama 2)

(i) Watoto wenyewe wamekunywa maziwa yao yote. (tendwa)

.....

.....

(ii) Pombe haramu inawafisha Wakenya wengi. (Badilisha kitenzi kiwe nomino)

.....

.....

(f) Tunga sentensi ukitumia kielezi cha jinsi cha nomino ifuatayo.

Ujerumani.

(alama 1)

.....

(g) Tambua aina za vitenzi katika sentensi zifuatazo:

(alama 3)

(i) Mjomba alikuwa uwanjani.

.....

.....

.....

(ii) Mjukuu wake ataweza kulima.

.....

(h))Taja aina ya kirai kilichopigwa mstari katika sentensi ifuatayo.

Walimu wa Kiswahili watawasili leo. (alama.1)

.....

(i)Kanusha

Wanafunzi watundu waliadhibiwa na kuaibishwa. (alama 2)

.....

.....

.....

(j)Nyambua vitenzi vituatavyo katika kauli ulizopewa. (alama 2)

(a)Oa (tendesha)

(b)Kokoa..... (tendata)

(k) Tunga sentensi mojamoya kubainisha: (alama 2)

(i)Kiwakilishi cha pekee chenye maana ya umilikaji.

.....

.....

(ii)Kiwakilishi cha nafsi kiambata.

.....

.....

(l)Andika kinyume; (alama 2)

Shangazi angeingia tungemsifu.

.....

.....

(m)Tunga sentensi mojamoya kudhihirisha matumizi ya (alama 2)

(a)“ka” ya mfululizo

.....
.....

(b)“a” ya wakati usiodhihirika

.....
.....

(n) Changanua sentensi ifuatayo kwa kutumia visanduku:

Rosamaria alipata kitabu chake kipyua kilichopotea jana.

(alama 4)

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.....
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.....

(o) Tambua na uainishe viwakilishi katika sentensi ifuatayo: Aliniitisha zawadi uliyokuwa umenituma
nimpelekee Hadija. (alama 2)

.....
.....
.....
.....

(p) (i) Kishazi ni nini?

(Alama 1)

.....
.....

(ii) Tunga sentensi iliyo na kishazi tegemezi

(alama 1)

.....
.....

(q)Tumia ‘o’ rejeshi tamati kuunganisha sentensi zifuatazo

(i) Jino linauma

(ii)Jino limeng'olewa

(alama2)

.....

.....

.....

(r) Tambua vivumishi na uonyesha ni vya aina gani .

Wanafunzi wawa hawa waliochelewa watafanya kazi ya sulubu kama adhabu.

(alama3)

.....

.....

.....

4.ISIMU JAMII (ALAMA 10)

Soma makala yafuatayo kisha ujibu maswali

USAJILI WA WATAHINI

Wizara ya Elimu, Sayansi na Teknolojia imewaagiza walimu wakuu wa shule za msingi na sekondari wabandike hadharani orodha ya watahiniwa wote waliosajiliwa ili kuwawezesha wazazi na walimu kuthibitisha maelezo yote ya usajili.

Kwenye taarifa aliyoitoa kwa vyombo vya habari, Waziri Dkt Matiang'i alitoa wito kwa wazazi na wanafunzi watakaofanya mitihani ya kitaifa mwaka huu wa 2017 wahakikishe kuwa watahiniwa wamesajiliwa vilivyo. Aliwataka wathibitishe usajili wao katika tarakilishi na kutia saini nakala hizo kama inavyohitajika.

(a)Tambua sajili ya makala haya.

(alama 2)

.....

(b)Eleza sifa nane za kimsingi za sajili hii.

(alama 8)

KCSE MOCKS SET 2

102/3

KISWAHILI

Karatasiya 3

FASIHI

Muda: Saa 2½

Karatasi hii ina kursa nne zilizopigwa chapa.

SEHEMU YA A: RIWAYA
Assumpata K. Matei: Chozi la Heri

LAZIMA

1. “Haiwezekani! Hili haliwekazi! Itakuwa kama kile kisa cha yule kiongozi wa kiimla wa kike”

(a) Eleza muktadha wa dondoo hili. (alama 4)

(b) Eleza kwa kifupi yaliyowapata wahafidhina baada ya usemi huu. (alama.4)

(c) Riwaya ya chozi la heri inaonyesha maovu yanayotamalaki katika jamii. Thibitisha. (alama 12)

SEHEMU B: TAMTHILIA
Kigogo. Pauline Kea.

Jibu swali la 2 au la 3

2. Tatizo la uongozi katika bara la Afrika ni kikwazo kikubwa cha maendeleo. Kwa kurejelea matukio kwenye tamthilia ya kigogo, jadili ukweli wa kauli hii (alama 20)

AU

3. “Mtalipa kila tone la damu mlilomwaga sagamoyo ;wewe na watu wako.”

a) Eleza muktadha wa maneno haya (alama 4)

- (c) Eleza sifa za mzungumzaji (alama 4)

- (b) Thibitisha kwa kutoa mifano kuwa maisha ya anayezungumziwa yametawaliwa na dhuluma. (alama 12)

SEHEMU YA C: HADITHI FUPI

4. Kwa kurejelea hadithi ya mapenzi ya kifaurongo, onyesha jinsi jamii imegawanyika kitabaka kuegemea.

- i. Kielimu
- ii. Kikazi
- iii. Kiuchumi (alama 20)

SEHEMU D: SHAIRI

WASIA

5. Huno wakati mufti, vijana nawausia
Msije juta laiti, mkamba sikuwambia
Si hayati si mamati, vijana hino dunia
Uonapo vyang'aria, tahadhari vitakula
- Japo aula kushufu, na machoni vyavutia
Dunia watu dhaifu, yaugua nasikia
Vijana nawasarifu, falau mkisikia
Uonapo vyang'aria, tahadhari vitakula.
- Jepusheni na zinaa, mlale penye sheria
Msije andama baa, makaa kujipalia
Jepusheni na zinaa, madhara kukadiria
Uonapo vyang'aria, tahadhari vitakula.
- Ngawa waone wazuri, nadhifu kukuvalia
Wajimwaie uturi, na mapoda kumichia
Si mlango nyumba nzuri, ngia ndani shuhudia
Uonapo vyang'aria, tahadhari vitakula.
- Wawapi leo madume, anasa walopapia?

Wamepita ja umeme, leo yao sitoria
Shime enyi wana shime, bora kumcha Jalia
Uonapo vyang'aria, tahadhari vitakula

Nambie faida gani, nambie ipi fidia
Upatayo hatimani, waja wakikufukua
Ila kufa kama nyani, kasoro yako mkia
Uonapo vyang'aria, tahadhari vitakula.

Vyatiririka tariri, vina vyanikubalia
Alo bora mshairi, pa tamu humalizia
Nahitimisha shairi, dua ninawapigia
Uonapo vyang'aria, tahadhari vitakula.

Ewe Mola mtukuka, si shaka wanisikia
Wakingie wanarika, na anasa za dunia
Amina wangu Rabuka, dua yangu naishia
Uonapo vyang'aria, tahadhari vitakula.

MASWALI:

- a) Ni ujumbe gani wanaopewa vijana kupitia shairi?. (alama 4)
- b) Bainisha tamathali mbili za usemi katika shairi hili. (alama 2)
- c) Eleza bahari ya shairi hili kwa kuzingatia vigezo vifuatavyo. (alama 2)
 - i) idadi ya vipande katika mshororo
 - ii) mpangilio wa vina katika beti.
- d) Eleza mbinu zozote mbili za kishairi zilizotumika katika shairi hili. (alama 2)
- e) Andika ubeti wa pili kwa lugha nathari (alama 4)
- f) Eleza toni ya shairi hili. (alama 1)
- g) Tambua: (alama 2)
 - i) Nafsi neni
 - ii) Nafsi nenewa
- h) Eleza umuhimu wa mbinu ya kimtindo iliyotumika katika ubeti wa tano. (alama 2)
- i) Eleza maana ya msamiati: 'aula' (alama 1)

SEHEMU YA E: FASIHI SIMULIZI

- 6. a) Eleza maana ya miviga. (alama.2)

- b) Eleza sifa tano za miviga. (alama.5)
- c) Miviga ina udhaifu gani. (alama.3)
- d) Fafanua umuhimu wa ngomezi katika jamii. (alama.6)
- e) Eleza vizingiti viwili vinavyokumba ngomezi. (alama.4)

Huu ni ndio ukurasa wa mwisho

KCSE MOCKS SET 2

MATHEMATICS

PAPER 1

SECTION A (50MARKS)

1. Evaluate Leaving your answer in fraction form(3 mks)

$$\frac{\frac{3}{5} \text{ of } 30 + 5\frac{5}{6} \div \frac{7}{12} - 2\frac{2}{3} \times 1\frac{1}{2}}{5\frac{5}{8} \times 1\frac{7}{9} - \frac{5}{9} \text{ of } 4\frac{4}{5} + \frac{14}{5} \div \frac{7}{10}}$$

.

2. Mr. Omondi left shs. 116,580 in his bank account to be shared between his wife, daughter and son in the ratio 1:2:3. His wife decided to divide her share equally between her daughter and son. Determine how much the son finally got. (3 mks)

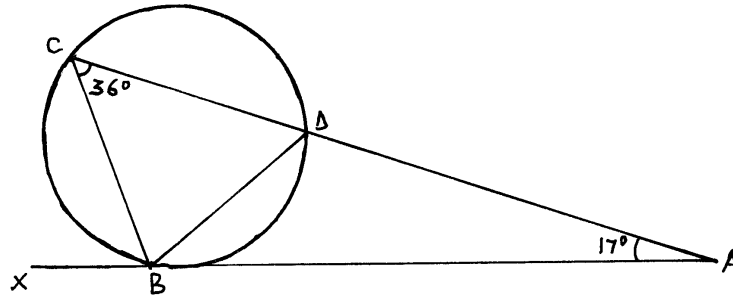
3. Use logarithms in all your steps to work out. (4mks)

$$\sqrt[3]{\frac{4968}{87.56 \div 6.258}}$$

4. Sankale walks for $2\frac{1}{2}$ hours in the morning at x km/hr and for $\frac{1}{2}x$ hours in the afternoon at 6 km/hr. This makes $38\frac{1}{2}$ km altogether. How far did she walk in the morning. (3mks)

5. Given that $\cos \theta = \frac{15}{17}$ and $270^\circ \leq \theta \leq 360^\circ$. Find without using tables the values of $\sin \theta$ and $\tan \theta$. (3mks)

6. On the figure below ABX is a tangent, Angle CAB = 17° and Angle ACB = 36° . Calculate angle CBX and angle DBC. (3mks)



7. What is the sum of the roots of the equation. (3mks)

$$\frac{x^2 + 3}{x} = 4$$

8. Sonko is a real estate agent who is entitled to a commission on all properties bought through him. During a certain month he sold 2 mansions at sh. 2.54 million each, 4 flats at sh. 582,000 each and 5 bungalows at sh. 354,000 each. If he was paid a total commission of sh. 458,900. Calculate the percentage rate of commission he was paid.

(4mks)

9. It takes 20 men 10 days to lay 300 metres of pipes. Find how many days it would take 15 men to lay 270 metres of pipes working at the same rate. (3mks)
10. A cylindrical tank of diameter 1.4m and height 1.2m is one-quarter full of water. This water is transferred to an empty rectangular container measuring 1.2m long and 70cm wide. Calculate the height of the water in the container in centimeters. (3mks)
11. Give the integral values of x which satisfies the following inequalities.
 $4 < 3x - 2$; $15 - 2x > 4$. (4mks)

12. The average mark scored by the first 27 students in a mathematics test is 52. The average mark scored by the remaining 37 is 58. Calculate the mean mark for the whole class.
(4mks)
13. Nyambura bought 3 skirts and 2 sweaters at a total cost sh. 1575. If he had bought 2 shirts and 3 sweaters he would have spent sh. 225 more. Find the cost of 5 skirts and 2 sweaters.
(4mks)
14. The straight line whose equation is $2y = 3x + 6$ meets the x-axis and the y-axis at P and Q respectively. Write down the coordinates of P and Q.
(3mks)
15. Nairobi and Eldoret are 351 km apart. A bus leaves Nairobi towards Eldoret at an average speed of 66km/h. At the same time a car leaves Eldoret traveling at an average speed of 104km/hr towards Nairobi. Along the way the car stopped for 10 minutes to repair a puncture, then resumed the

journey traveling at the same average speed. How far from Nairobi did they meet.

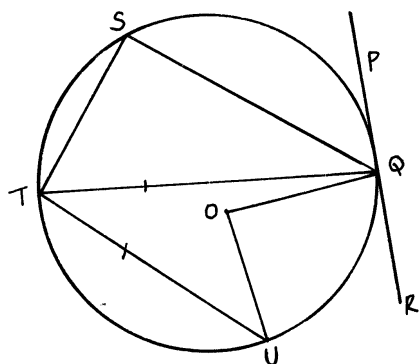
(4mks)

16. A number P is divided by 12,15 and 18. In each case the remainder is 5. Find the smallest value of P. (3mks)

Section II (50 Marks)

Answer ANY FIVE questions

17. In the figure below, O is the centre of the circle. PQR is a tangent to the circle at Q, Angle PQS = 28° , angle UTQ = 54° and UT = TQ. Giving reasons, determine the size of



a) Angle STQ (2 mks)

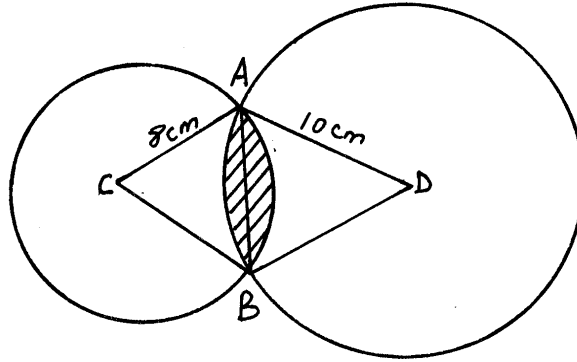
b) Angle TQU (2 mks)

c) Angle TQS (2 mks)

d) Reflex angle UOQ (2 mks)

e) Angle TQR (2 mks)

18. The figure below shows two circles, centres 'C' and 'D' of radii 8cm and 10cm respectively. The two circles subtend by angles of θ and β respectively at their centres and intersect at A and B as shown.



a) Given that the area of triangle $ACB = 30.07\text{cm}^2$ and that of triangle $ADB = 43.30\text{cm}^3$. Calculate the size of the

(i) angle marked θ (2 mks)

(ii) angle marked β (2 mks)

b) Calculate to two decimal places the area of

(i) Sectors ACB (2 mks)

(ii) Sector ADB (2 mks)

(iii) The shaded region

(2 mks)

19. a) Two ships Q and R are sailing towards port P. At 1144 hours ship Q is exactly 120km on a bearing of 030^0 from P and ship R is 50km on a bearing of 300^0 from P. At this instant, ship R develops engine trouble and cannot continue with the journey. Ship Q receives the distress signal from ship R and has to change course and steam straight towards ship R at 50km/h. Without using a scale drawing, calculate the time of day ship Q reaches ship R.

(5 mks)

- b) A particle moves along a straight line OA such that t seconds after it is at 'O'; its velocity is v m/s where $v=qt - 2t^2$ and q is a constant. At the point P, $t = 4$ and the particle is momentarily at rest. Calculate;

(i) The value of q

(1 mk)

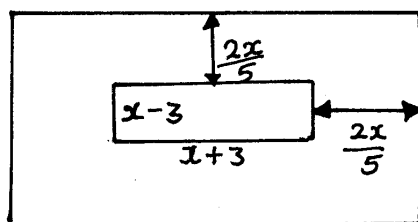
(ii) The distance OP

(2 mks)

(iii) The acceleration when $t = 1\frac{1}{2}$ seconds.

(2 mks)

20. The figure below represents the floor of a dancing hall with a carpeted margin all around of $\frac{2x}{5}$ m wide leaving a dancing space of $(x-3)$ m by $(x+3)$ m



- a) If the total area of the entire room is 315m^2 , calculate the value of x (4 mks)

- b) Hence calculate the area of the carpeted margin.

(2 mks)

- c) If the carpet cost shs. 750 per m^2 . Calculate the total cost of the sealed margin.

(2 mks)

21. a) The points A^I B^I C^I are the images of $A(4,1)$ $B(0,2)$ and $C(-2,4)$ respectively under a transformation represented by matrix $M = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$. Write down the co-ordinates of A^I B^I C^I

(3 mks)

- b) A^{II} B^{II} C^{II} are the images of A^I B^I C^I under another transformation whose matrix is

$N = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$. Write down the co-ordinates of A^{II} B^{II} C^{II} . (3 mks)

- c) Transformation M followed by N can be replaced by a single transformation P . Determine the matrix for P . (2 mks)

d) Hence determine the inverse of matrix P. (2 mks)

22. Draw triangle PQR with vertices P(2,3) Q(1,2) and R(4,1) and triangle $P'' Q'' R''$ with vertices $P'' (-2,3)$ $Q'' (-1,2)$ $R'' (-4,1)$ on the same axes. (2 mks)

(i) Describe fully a single transformation which maps triangle PQR onto triangle $P'' Q'' R''$. (2 mks)

(ii) On the same plane, draw triangle $P^1 Q^1 R^1$ the image of triangle PQR, under reflection in line $y = -x$ (2 mks)

(iii) Describe fully a single transformation which maps triangle $P^I Q^I R^I$ onto $P^{II} Q^{II} R^{II}$
(2 mks)

(iv) Draw triangle $P^{III} Q^{III} R^{III}$ such that it can be mapped onto triangle PQR by a positive quarter turn about the origin (0,0)
(2 mks)

23. a) Draw a graph of $y = 8 - 10x - 3x^2$ for $-5 \leq x \leq 3$ (5 mks)

b) On the same axes, draw the line $y = 2x + 1$ and hence find;

(i) The roots of $8 - 10x - 3x^2 = 2x + 1$ (2 mks)

(ii) A quadratic equation with roots in b(i) above. (1 mk)

c) By including a suitable straight line, use your graph to solve $3x^2 + 12x - 11 = 0$ (2 mks)

24. Using a ruler and a pair of compasses only. Construct a parallelogram ABCD such that $AB = 8\text{cm}$ diagonal $AC = 12\text{cm}$ and angle $BAC = 22.5^\circ$ (4 mks)

- a) Measure (i) The diagonal BD (1 mk)
- (ii) The angle ABC (1 mk)
- b) Draw the circumcircle of triangle ABC (2 mks)
- c) Calculate the area of the circle drawn. (2 mks)

KCSE MOCKS SET 2

121/2

Mathematics

Paper 2

Time: 2 ½ hours

1. Evaluate without using Mathematical tables or a calculator.

(3mks)

$$2\log 5 - \frac{1}{2}\log 6 + 2\log 40$$

2. Solve for x given that the following is a singular matrix

(2mks)

$$\begin{pmatrix} 1 & 2 \\ x & x-3 \end{pmatrix}$$

3. Make d the subject of the formula.

(3mks)

$$a^2 = \sqrt{\frac{1+d^2}{b^2}} - \frac{b}{3}$$

4. Simplify $\frac{3}{\sqrt{7}-2} + \frac{1}{\sqrt{7}}$ leaving your answer in the form $a + b\sqrt{c}$, where a, b and c are rational numbers.
(3mks)
5. Calculate the percentage error in the volume of a cone whose radius is 9.0cm and slant length 15.0cm.
(3mks)
6. A quantity A is partly constant and partly varies inversely as a quantity B. Given that A = -10 when B = 2.5 and A = 10 when B = 1.25, find the value of A when B = 1.5.
(4mks)

7. The table below shows corresponding values of x and y for a certain curve.

y	1.0	1.2	1.4	1.6	1.8	2.0	2.2
x	6.5	6.2	5.2	4.3	4.0	2.6	2.4

Using 3 strips and mid-ordinate rule, estimate the area between the curve x axis, the line $x = 1$ and $x = 2.2$.

(2mks)

8. 14 people can build 10 huts in 30 days. Find the number of people working at the same rate that will build 18 similar huts in 27 days.

(3mks)

9. The coordinates of two airports M and N are (60°N , 35°W) and (60°N , 15°E) respectively. Calculate;

(i) The longitude difference.

(1mk)

(ii) the shortest time an aeroplane whose speed is 250 knots will take to fly from M to N along a circle of latitude.

(2mks)

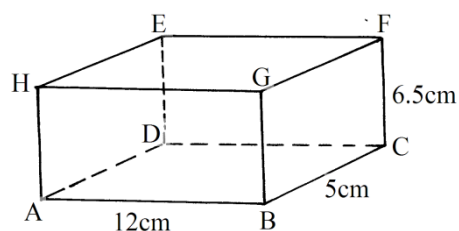
10. (a) Expand $(x - 0.2)^5$ in ascending powers of x .

(2mks)

- (b) Use your expansion up to the fourth term to evaluate 9.8^5 .

(2mks)

11. The figure below is a cuboid ABCDEFGH. $AB = 12\text{cm}$, $BC = 5\text{cm}$ and $CF = 6.5\text{cm}$.



- (a) State the projection of AF on the plane ABCD.

(1mk)

- (b) Calculate the angle between AF and the plane ABCD correct to 2 decimal places.

(3mks)

12. Show that $\frac{\sin x(\cos x + 1)}{\cos x} = \sin x + \tan x$.

(3mks)

13. The mid-point of AB is (1,-1.5, 2) and the position vector of a point A is $-\hat{i} + \hat{j}$. Find the

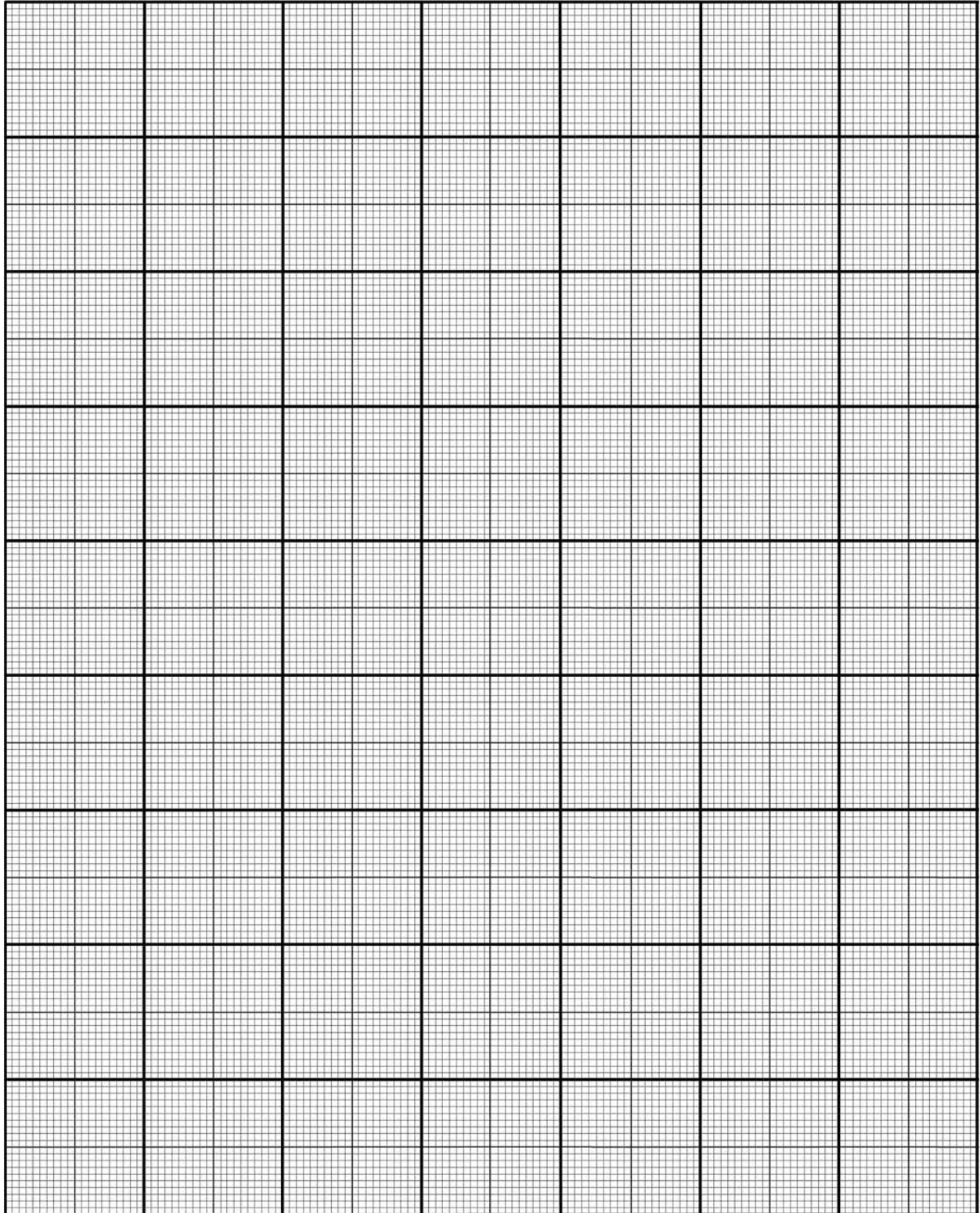
magnitude of $\vec{AB}$ where O is the origin.

(3mks)

14. Draw a line of best fit for the graph of y against x using the values in the table below.

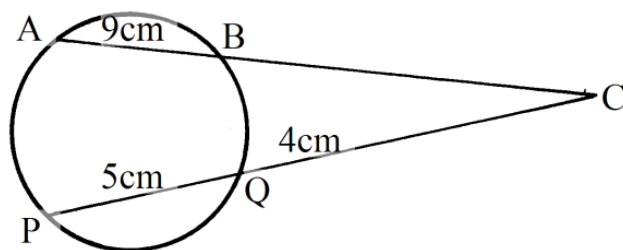
Hence determine the equation connecting y and x.

x	0.4	1.0	1.4	2.0	2.5
y	0.5	1.0	1.2	1.5	2.0



15. A coffee dealer mixes two brands of coffee, x and y to obtain 40kg of the mixture worth Ksh. 2,600. If brand x is valued at Ksh. 70 per kg and brand y is valued at Ksh. 55 per kg. Calculate the ratio in its simplest form in which brands x and y are mixed.
(4mks)

16. The figure below shows a circle centre O. AB and PQ are chords intersecting externally at a point C. AB = 9cm, PQ = 5cm and QC = 4cm. Find the length of BC.
(3mks)



SECTION II (50 MARKS)

Answer only five questions in this section

17. An examination involves a written and a practical test. The probability that a candidate passes the written test is $\frac{6}{11}$. If a candidate passes a written test then the probability of passing the practical test is $\frac{3}{5}$ otherwise it would be $\frac{2}{7}$.

(a) Illustrate this information on a tree diagram (2mks)

(b) Determine the probability that a candidate is awarded

(c) (i) credit for passing both tests (2mks)

(ii) pass for passing the written test (2mks)

(iii) retake for passing the test (2mks)

(iv) Fail for not passing the test (2mks)

18. . The relationship between the variables a and y is believed to be $y = a/x + bx$. Where a and b are constants. The table below shows corresponding values of x and y

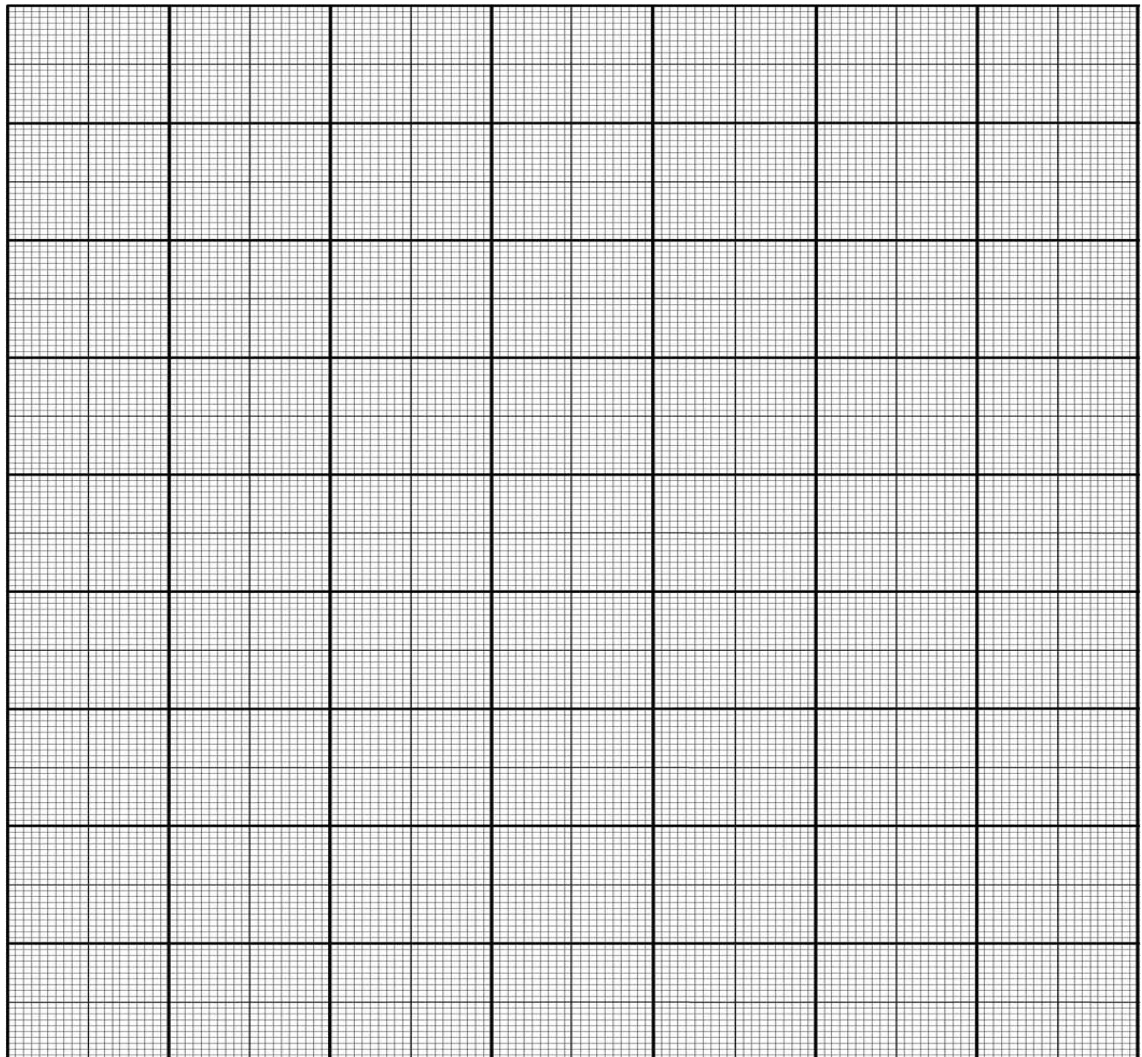
x	1	2	3	4	5
y	5.00	7.00	9.67	12.50	15.40

- (a) Write the relationship in the form of $y = mx +$

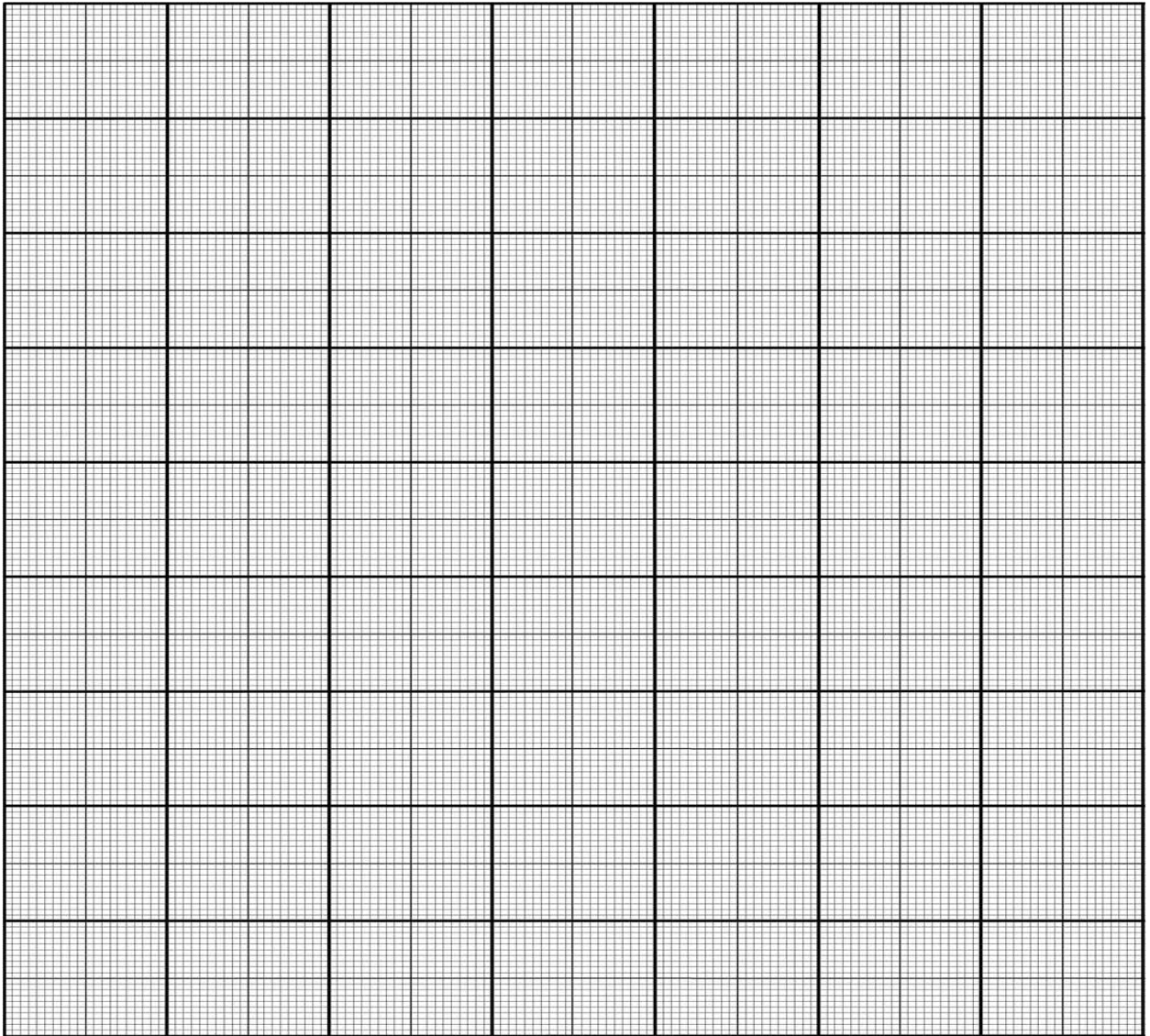
c

(1mk)

- (b) By drawing a suitable straight line graph estimate the values of a and b
(7mks)



- (c) Find the value of y when $x = 1000$
(2mks)
19. The vertices of triangle ABC are $a(3,1)$ $B(0,2)$ and $c(2,-1)$
- (a) $A'B'C'$ is the image of ABC under reflection on the line $y = x$. Draw $A'B'C'$ on the grid provided hence state the co-ordinates of its vertices
(3mks)
- (b) $A''B''C''$ is the image of $A'B'C'$ under positive quarter turn about the origin. Draw triangle $A''B''C''$ and state the co-ordinates of its vertices.
(3mks)
- (c) $A'''B'''C'''$ is the image of triangle ABC under shear matrix, y axis invariant and linear scale factor 3. Write down the shear matrix hence find the co-ordinates of the vertices of triangle $A'''B'''C'''$ (1mk)



20. Two points P and Q are found on the earth's surface the position of P is ($52^{\circ}\text{S}, 66^{\circ}\text{W}$) and Q ($52^{\circ}\text{S}, 144^{\circ}\text{E}$). Taking earth's radius as 6370km,

(a) Find the difference in longitude between the two points P and Q
(1mk)

(b) (i) calculate the shortest distance between points P and q along (i) the latitude in km
(1mk)

(ii) The longitude in Km
(4mks)

(

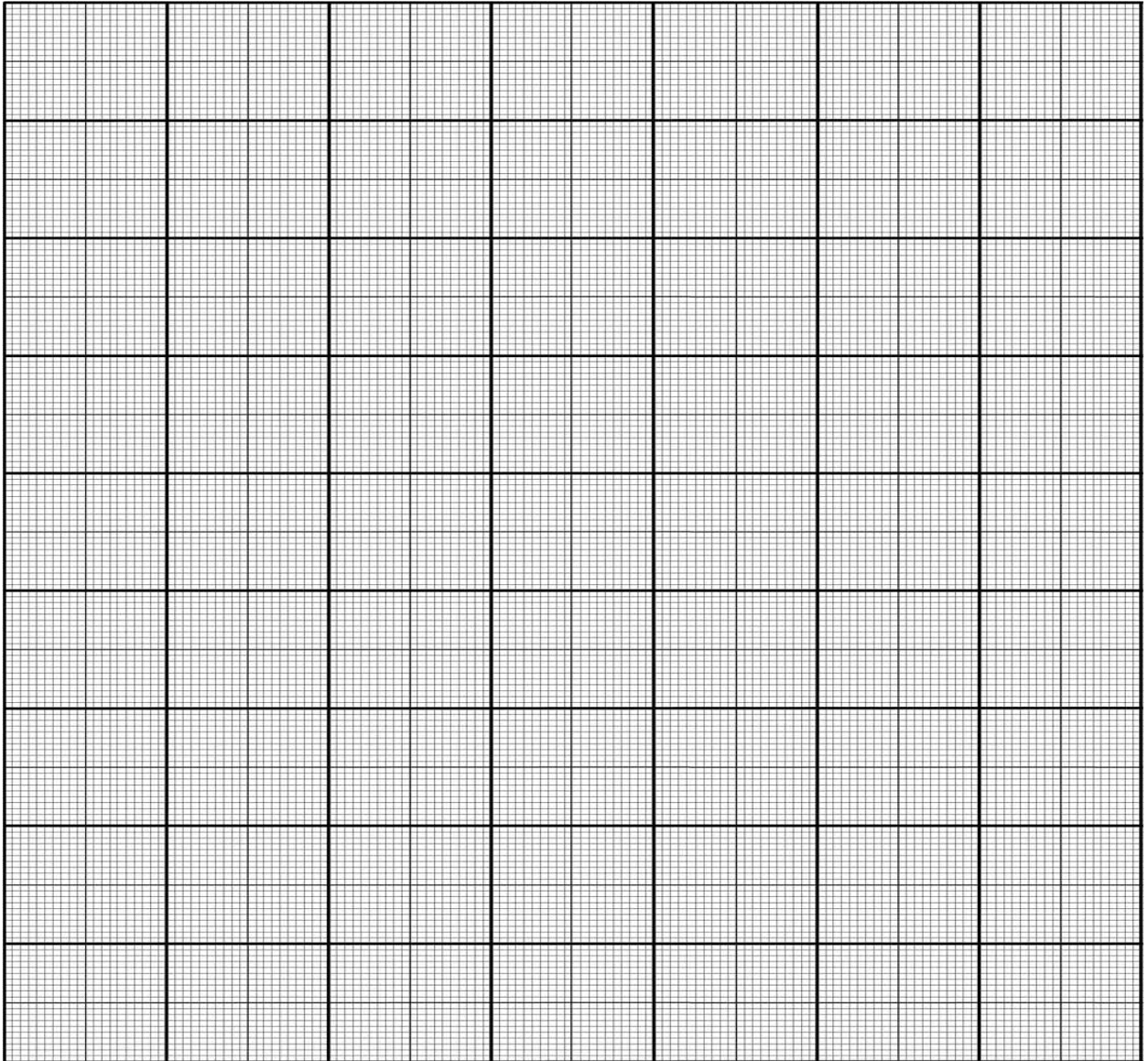
(d) A plane travelling at 800km/hr leaves point P At 10.00am and sails through south pole to point q. Find the local time the plane arrives at point Q to the nearest minute.
(4mks)

21. A company has two types of machines A and B for juice production. Type A machine can produce 800 litres per day while type B machine can produce 1600 litres per day. Type A machine needs four operators and type B needs seven operators. At least 800 litres must be produced daily and the total number of operators should not exceed 41. There should be two or more machines of each type.

Letting x be the number of machines of type A and y for type B.

(a) Form all inequalities in x and y to represent the above information
(4mks)

- (b) On the grid provided below, draw the inequalities and shade the unwanted region
(4mks)



- (c) Use the graph I (b) above to determine the least number of operators required for the maximum possible production.
(2mk)

22. Using a ruler and a compass only, construct a triangle ABC such that $AB = 6.8 \text{ cm}$, $BC = 5.6 \text{ cm}$ and angle $ABC = 37 \frac{1}{2}^\circ$
(3mks)

(b) Locate the :

- (i) Locus P such that angle $APB = \text{angle } ACB$
(3mks)

- (ii) Locus Q such that Q is equidistant to points A and B
(2mks)

- (iii) Locus R such that R is equidistant to lines AB and AC
(2mks)

23. The distance S meters from a fixed point O , covered by a particle after t seconds is given by the equation $S = t^3 - 6t^2 + 9t + 5$

(a) calculate the gradient of the curve at $t = 0.5$ seconds
(3mks)

(b) Determine the values of S at the turning points of the curve (3mks)

(c) Sketch the curve in the space provided.
(4mks)

24. The table below shows the distribution of marks obtained by 50 students

Marks	45-49	50-54	55-59	60-64	65-69	70-74	75-79
No of students	3	9	13	15	5	4	1

(a) Calculate the mean using asuitable assumed mean
(3mks)

(

(b) calculate the variance
(3mks)

(

(c) calculate standard deviation
(1mk)

(

(d)

(e) If 30 students were to pass ,calculator the pass mark (give your answer to nearest whole mark)
(3mks)

KCSE MOCKS SET 2

PHYSICS PAPER 1

INSTRUCTION: ANSWER ALL THE QUESTIONS IN THE SPACES PROVIDED.

SECTION A (25 MARKS)

Answer ALL the questions in this section in the spaces provided.

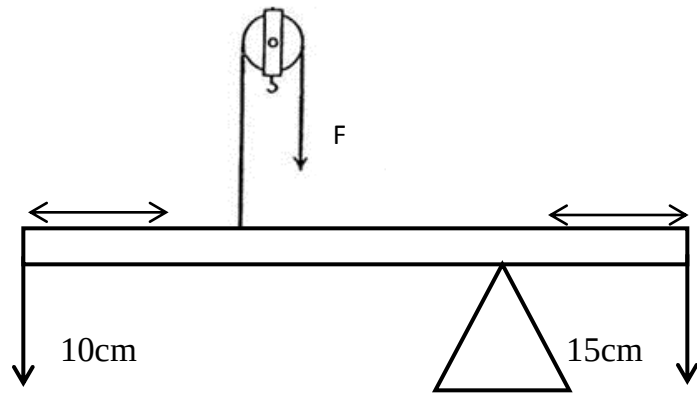
1. A student measured the radius of a cylindrical container and gave it as 200NM.
What instrument did the student use? (1mk)
2. The level of water in a burette is 25cm³. If 40 drops of average volume 0.05cm³ are run out of the burette, what would be the new level? (1mk)
3. Water and milk are mixed in the ratio 4:1 respectively. If the density of water is 1g/cm³ and that of milk is 1.2gcm⁻³, find the mass in grams, of 2.5 litres of the mixture. (3mks)

4. The weight of a sack of sawdust on earth is found to be equal to the weight of a sack on the surface of the moon. Explain this.
(1mk)

5. A needle floats on pure water, but sinks when a detergent is added to the water. Explain this observation. (1mk)

6. A hole of area 2.0cm^2 at the bottom of a tank of depth 2m is closed with a stopper. Determine the force on the stopper when the tank is filled with water. (Density of water = 1000kgm^{-3} and acceleration due to gravity is 10N/kg). (3mks)

7. A uniform plank of weight 10N and length 50cm is pivoted at a point X along its length. Weights of 15N and 20N and force F act on it as shown below. If the system is in equilibrium, calculate the value of F. (2mks)



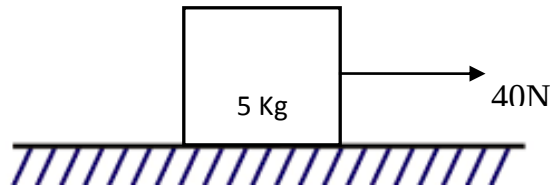
15N

X

20N

8. A stone floor feels cold to the feet, but a woolen carpet on the same floor feels warm. Explain this. (1mk)

9. The figure below shows a force of 40N acting on a body mass 5kg. The frictional force of the body is 5N. Determine the acceleration of the body. (3mks)



10. An ungraduated thermometer attached to a centimetre scale reads 7.5cm in melting ice at 0°C and 23.5cm in steam at 100°C . The same thermometer reads 5.5cm when placed in a freezing mixture. What is the temperature of the freezing mixture. (3mks)

10. A turntable of radius 8cm is rotating at 66 revolutions per second. Determine its linear speed at a point on its circumference. (3mks)

11. A solid displaces 5.5cm^3 of paraffin when floating and 20.0cm^3 when fully immersed. If the density of paraffin is 0.8g/cm^3 , calculate the density of the solid.
(3mks)

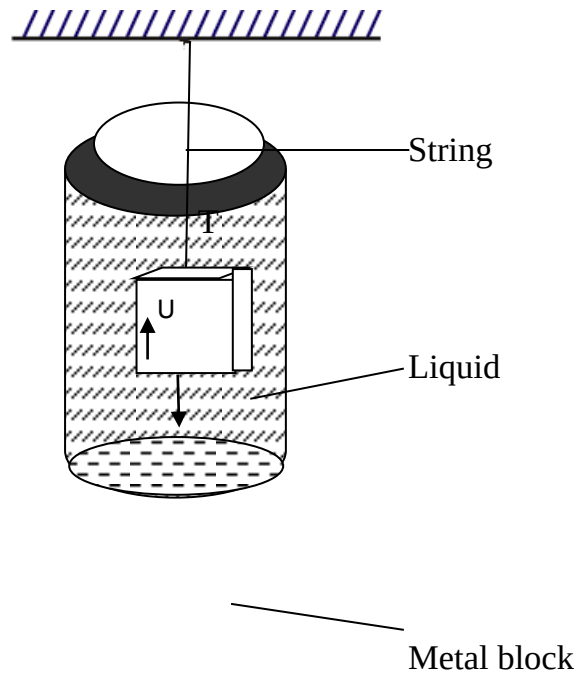
12. A tourist wanted to have a warm bath at 50°C . She had 5.0kg of water in a basin at 80°C . What mass of cold water at 30°C must she add to the hot water to have her bath of choice?

(Neglect heat loss and take specific heat capacity of water as 4200J/kg/K).
(3mks)

SECTION B

14(a) State the law of flotation.

(b) The following set up shows a metal block of density 10500kg/m^3 and dimension $30\text{cm} \times 20\text{cm} \times 20\text{cm}$, suspended inside a liquid of density 1200kg/m^3 . The block is held by a string attached to a point above the liquid. The three forces acting on the block are: Tension T on the string, weight W of the block and upthrust U due to the liquid.



i) Write an expression relating T , W and U if the block is in equilibrium inside the liquid.

(1mk)

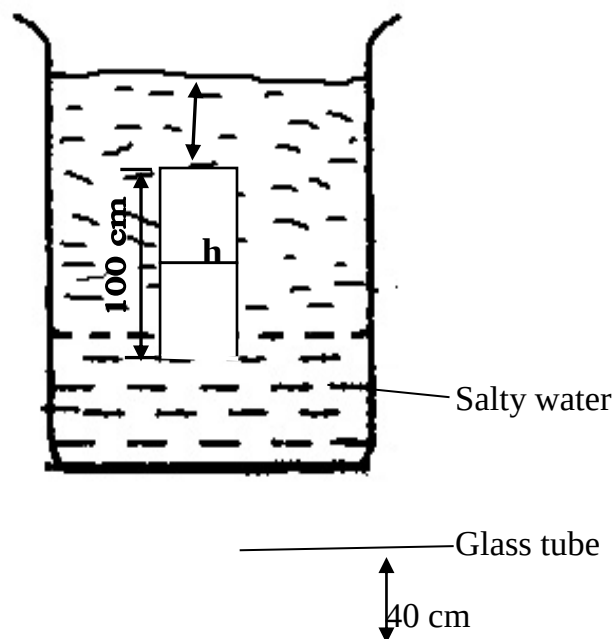
ii) Determine the weight, W , of the block.

(2mks)

iii) Determine the weight of the liquid displaced by the fully submerged block.
(2mks)

iv) Determine tension T in the string.
(1mk)

15. The set up below was used by a student to investigate the variation of volume and pressure for a fixed mass of a gas.



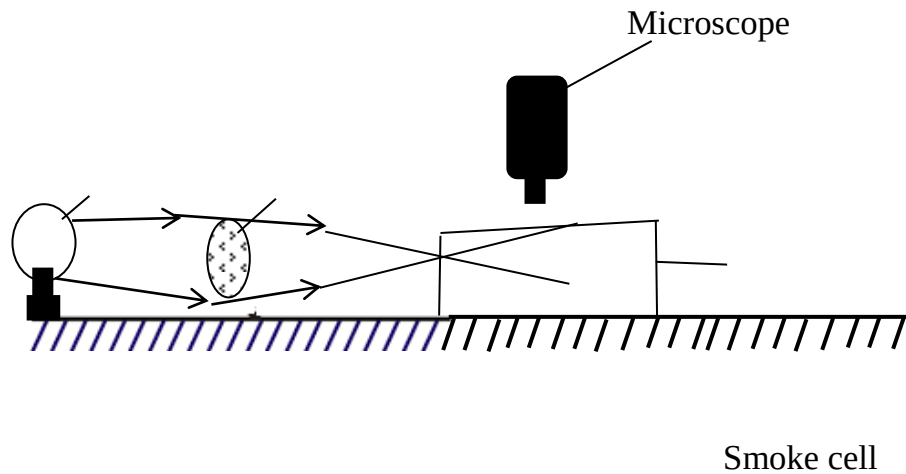
The student lowered a 10m long glass tube of uniform cross sectional area A vertically, mouth downwards into a container of salty water of density 1200 kg/m^3 . Water rose to a level of 40cm inside the glass tube. The experiment was performed on a day when the atmospheric pressure was 760mmHg and at a constant room temperature. If the density of mercury is 13600 kg/m^3 determine:

i) The atmospheric pressure in N/m^2 . (2mks)

ii) The height, h , of the tube below the water surface. (2mks)

(b) State the law used in working out the answer to question (ii) above. (1mk)

16. Brownian motion can be studied using the apparatus shown in the following figure. To observe the motion, some smoke is enclosed in the smoke cell and then observed through the microscope.



(a) Explain the role of the smoke particles, the lens and the microscope in the experiment. (3mks)

(b) State and explain the nature of the motion observed in the smoke particles. (2mks)

(c) What will be observed about the motion of the smoke particles if the temperature in the smoke cell is raised. (1mk)

(d) In an experiment to estimate the thickness of an oil molecule, an oil drop of diameter 0.1cm spread onto a circular patch of diameter 10cm.

i) Determine the volume of the oil drop. (2mks)

ii) Calculate the area covered by the oil patch. (2mks)

iii) Determine the thickness of the oil molecule. (2mks)

iv) State two assumptions made in (d) (iii) above. (1mk)

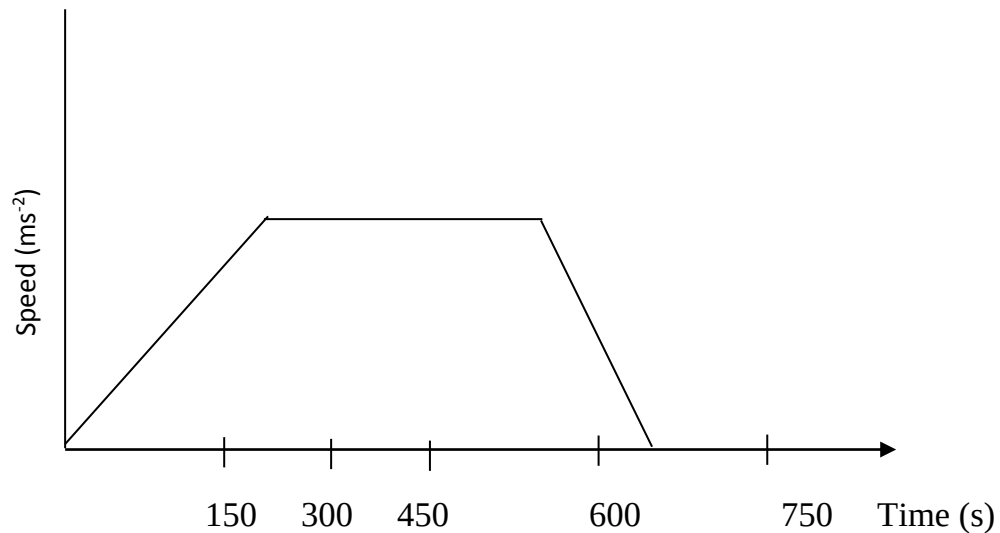
v) State two possible sources of errors in this experiment. (1mk)

17. A stone is thrown vertically upwards from the top of a tower 30m high, with an initial velocity of 20m/s. Determine:

i) The time it takes to reach maximum height. (2mks)

ii) The total time which elapses before it hits the ground. (2mks)

- (b) The diagram below shows a speed time graph for train traveling between two stations. The train starts from rest and accelerates uniformly for 150 seconds. It then travels at a constant speed for 300 seconds and finally decelerates uniformly to rest in 200 seconds.



Given that the distance between the two stations is 10450m, Calculate:

i) the maximum speed attained by the train in km/h. (3mks)

ii) the acceleration for the first 150 seconds. (2mks)

iii) the distance traveled during the last 100 seconds. (2mks)

iv) the time taken to travel the first half of the journey. (3mks)

18(a) Define the term work. (1mk)

(b) The table below shows the values of extension of spiral when various forces are applied to it.

Force F(N)	0	1.0	2.0	3.0	4.0	5.0	6.0
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Extension e(cm)	0	0.8	1.5	2.3	3.1	3.8	4.6
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i) Plot a graph of force (y-axis) against extension (x axis). (5mks)

ii) Determine the work done in stretching the spring by 4.0cm.. (3mks)

(c) A pump can raise 120kg of water to a height of 10.0m every minutes.

i) What is the power output of the pump. (3mks)

ii) If the efficiency of the pump is 80%, what power must be supplied to the pump?
(3mks)

19(a) The following table shows the values of the square of velocity v^2 , and distance moved

for uniformly-accelerated car. Use the information in the table to answer the questions

that follow.

Distance $s(\text{m})$	0	5	10	15	20	25
Squared velocity $v^2 (\text{m}^2/\text{s}^2)$	0	20	40	60	80	100

i) Plot a graph of the square of velocity, v^2 (y axis) against the distance, s). (5mks)

ii) From the graph, determine the acceleration of the car. (3mks)

(b) A body moving at 50ms^{-1} decelerates uniformly at 2ms^{-2} until it comes to rest.
What distance does it cover from the time it starts to decelerates to the time it comes to

rest? (4mks)

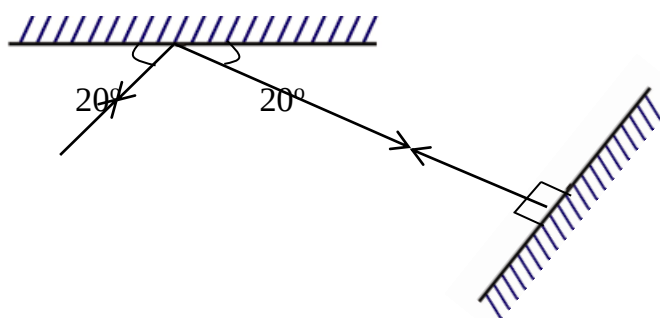
KCSE MOCKS SET 3

PHYSICS PAPER 2

SECTION A (25 MARKS)

Answer ALL the questions in this section in the spaces provided.

1. The following diagram shows the path of a ray of light after striking two mirrors at an angle.



Determine the angle between the two mirrors.

(2mks)

2. A rod rubbed with a duster is observed to attract tiny pieces of paper. However, as soon as

the papers touch the rod, they are repelled. Explain this observation. (2mks)

3. State two factors that determine the capacitance of a parallel-plate capacitor. (2mks)

4. State two ways in which polarization reduces the p.d. across a simple cell. (2mks)

5. Using the domain theory of magnetism, explain how heating a magnet weakens its magnet.
(2mks)

6. Explain why the coil in an electric motor is usually wound on a laminated soft iron core.
(2mks)

7. The figure below shows a wave on a string.

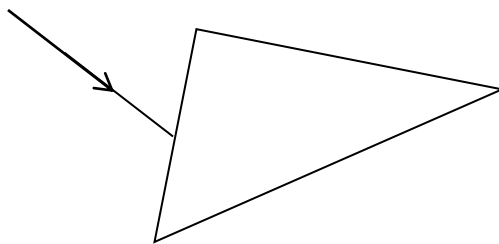




Determine the frequency of the wave. (3mks)

8. State the difference between sound waves and electromagnetic waves. (1mk)

9. The diagram below shows a ray of light incident on a glass prism at an angle.

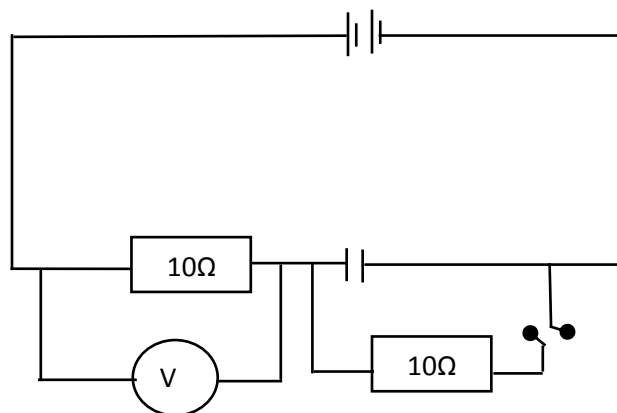


Complete the diagram to show the ray as it emerges from the other side. (2mks)

10. The diagram below shows a circuit containing a battery, two resistors, a capacitor and voltmeter.

Determine the reading on the voltmeter before the switch is closed and after the switch is

closed. (3mks)



11. Define the term 'electron volt' (1mk)

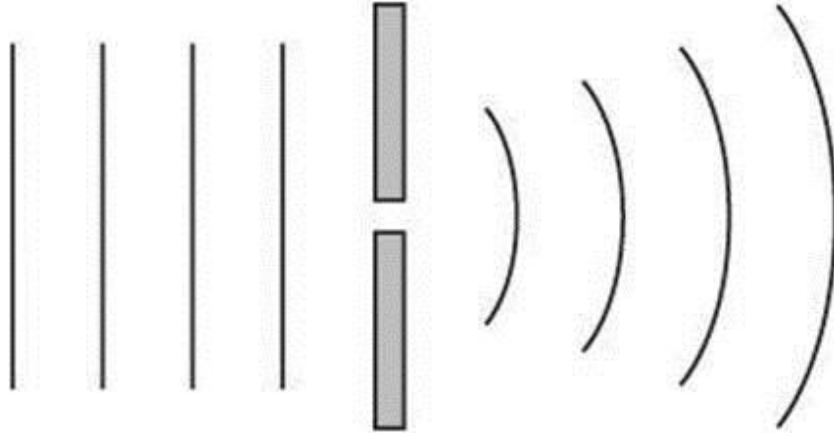
12. Sketch a circuit for a p-n diode in reverse bias. (1mks)

13. You are provided with three resistors of values 1 , 3 and 4 . Draw a circuit diagram to show the three resistors can be arranged so as to produce a resultant resistance of 3.8sl. (2mks)

SECTION B (50 MKS)

14(a) Define the term ‘diffraction’ as applied in waves. (1mk)

(b) The diagram below shows wave fronts before and after passing an opening.



State what would be observed on the pattern after passing the opening if:

i) the gap was made smaller. (1mk)

ii) the wavelength was made very large. (1mk)

(c) When a metre rule was placed in a ripple tank where straight waves being produced by vibrator, it was noted that the distance between 12 successive dark lines (crest) was 30cm. the frequency of the b=vibrator was 20HZ. Determine.

i) the wavelength of the waves in the ripple tank. (2mks)

ii) the periodic time of the waves. (2mks)

iii) the velocity of the waves over the water surface. (3mks)

15(a) Define the term “supersonic speed” as applied in sound waves. (1mk)

(b) In an experiment to determine the speed of sound in air, a drum at a point 150m from a vertical wall was struck at varying frequencies while listening to the echo. The echo coincided with the sound from the drum at a time when 20 successive strikes were made within a time of 18.5s.

i) Determine the time taken for the echo to be heard. (2mks)

ii) Determine the speed of sound in air at the place. (2mks)

iii) What difference would you expect if the experiment was repeated on a colder day. (1mk)

(c) A boy strikes a railway line with a hammer. A railway worker 600m away hears two sounds, one from the railway line and the other from air. If the time interval between the two sounds is 1.65s and the speed of sound in air is 340 m/s determine the speed of sound in the railway line. (4mks)

16(a) Define the term ‘virtual image’ as applied in lenses. (2mks)

(b) You are provided with the following:

- A convex lens – A screen
- A metre rule - A candle

i) Sketch a diagram of a set-up that can be used in determine the focal length by lens – formula method using the apparatus.

(2mks)

ii) State the measurement that would be taken.

(2mks)

iii) Explain how those measurements would be used in determining the focal length of the lens.

(3mks)

(c) A small vertical object in placed 9cm from a converging lens of focal length 10cm.

Determine the nature of the image formed and the distance of the image from the lens.

(4mks)

17(a) State what is meant by half-life of a radioactive substance.

(1mk)

- (b) The activity of a sample of a radioactive substance was measured over a period of time. The table shows the results obtained.

Activity(disintegration/s)	680	567	473	395	276	160	112
Extension e(cm)	0	1	2	3	5	8	10

- i) Plot a graph of activity (y-axis) against time. (5mks)

- ii) Determine the half-life of the substance in days. (2mks)

- (c) The half life of a radioactive substance is 138 days. A sample of the substance has 8×10^{10} undecayed nuclei at time $t = 0$. How many undecayed nuclei will be left after 690 days? (3mks)

- 18(a) An x-ray tube produces X-rays whose wavelengths vary from 6.0×10^{-13} to 4.5×10^{-9} m.

Determine.

i) The range of its frequency of the x rays. (2mks)

ii) the highest energy of the x-rays. ($C=3.0 \times 10^8$ m/s and $h = 6.62 \times 10^{-34}$ js)
(3mks)

(b) A Surface with a threshold frequency of 5×10^{14} Hz is irradiated with a radiation of wavelength 6.0×10^{-7} m.

i) Define the term “threshold frequency”. (1mk)

ii) Determine the frequency of the radiation used. (2mks)

iii) Calculate the maximum kinetic energy of the photoelectrons emitted. (3mks)

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